

GREEN ROAD COMMUNITY CENTER HVAC RENOVATION

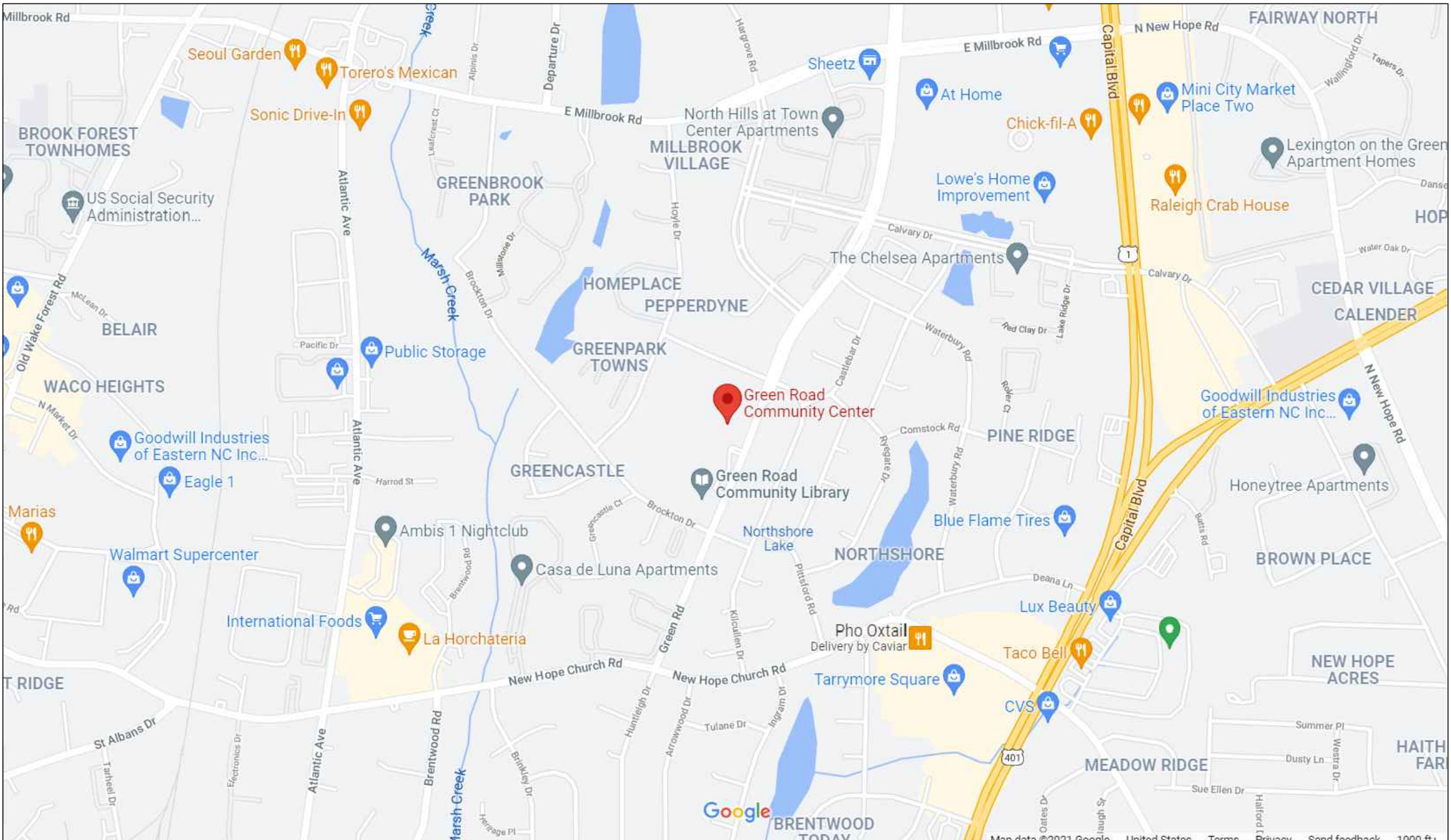
for: CITY OF RALEIGH, PARKS AND RECREATION DEPARTMENT

PROJECT NUMBER: 274-ESIFS-GRHVR-2021
RELEASED FOR CONSTRUCTION

DRAWING INDEX

C-1	COVER SHEET
C-2	APPENDIX B
M-1	MECHANICAL NOTES, LEGEND, SCHEDULE AND DETAILS
M-2	MECHANICAL DEMOLITION PLAN
M-3	BASE BID MECHANICAL RENOVATION PLAN
M-4	MECHANICAL RENOVATION UNIT PLAN
M-5	ALTERNATE BID MECHANICAL RENOVATION PLAN
M-6	CONTROL SCHEMATICS
M-7	MECHANICAL DETAILS
E-1	ELECTRICAL COVER
E-2	DEMO PLAN - ELECTRICAL
E-3	RENOVATION PLAN - ELECTRICAL
E-4	ALTERNATE RENOVATION PLAN
E-5	PANEL SCHEDULE
FA-1	FIRE ALARM NOTES AND LEGEND
FA-2	FIRE ALARM PLAN

VICINITY MAPS



CONTRACTOR RESPONSIBILITIES

- Contractor MUST visit job sites prior to submitting a bid. Bids submitted shall be considered verification of the contractor's knowledge of project conditions. The intent of the project as described herein (plans and/or specifications) is for the Contractor to provide the Owner with safe, code conforming, fully operational, and properly functioning equipment, systems and/or new construction as required to perform Owner's/User's task.
- Contractor shall be responsible for the means, methods, techniques, sequences, procedures and material supply for construction and installation, verification of dimensions at the site, the verification of existence and location of utility services (underground and above ground), and the verification of quantities.
- The Contractor shall be responsible for scheduling all trades work in complete coordination with the owner. Contractor shall meet with the owner to discuss scheduling prior to construction. Contractor shall make accommodations to minimize disruption to activities.
- Contractor shall, wherever possible, adhere to the drawings and specifications. Any variation from the drawings and specifications shall be approved in writing by the Engineer before Contractor proceeds with work. In an emergency, oral approval from the Engineer is sufficient but this must be followed with written approval. No claim for adjustment to the contract price shall be valid unless the procedure is followed.
- Contractor shall, under no circumstance, make any alteration to the existing building structure or utilities that will in any way jeopardize the structural stability or interrupt the building's operation without prior written permission from the Engineer.
- Successful bidder on the project (prior to construction) shall submit cut sheets/shop drawings for approval by the designer, detailing the devices and equipment he proposes to use.
- Contractor shall be responsible for removal of all debris and waste materials of construction off site unless specifically directed otherwise on the drawing or by the Engineer.
- Contractor shall be responsible for the acceptable closure and repair of all areas disturbed during construction, including, but not limited to, wall, floor, and ceiling penetrations, disturbed ceiling(s) and floor(s), fastening of supports, etc. Repair work shall utilize like materials where possible or materials compatible to the existing construction and shall restore the disturbed surface to original condition. Unless required otherwise, all repaired areas shall be finished to match adjacent existing surfaces, and exposed piping, duct work, conduit, and hanger assemblies shall be painted to match the existing features.
- Contractor shall consult with the Engineer prior to making any penetration or alteration of roof deck or existing roofing application, and shall obtain concurrence prior to, during, and upon completion of the work. All roof work must be performed by a licensed roof contractor and approved in writing by the Engineer. All damage to roof structure and waterproof membrane resulting from Contractor's activity shall be repaired (during the period of this contract, and as soon as possible) by the Contractor, at the expense of the Contractor, in a manner to meet any and all warranty that may be in effect.
- Building Utility Shutdown - The contractor shall notify the owner 10 working days before any unavoidable utility shutdown is to occur. ULOCO requires 48 hours notice. These include but not limited to such utilities as electrical, domestic, water, sewer, HVAC system, etc. Contractor shall keep down time to an absolute minimum.
- Contractor shall notify the Engineer immediately upon encountering any suspected asbestos product. Any removal must be coordinated through the Engineer's Office by approved contractors. The owner is not responsible for compensation due to delays for asbestos removal.
- Contractor shall be responsible for the restoration of all landscape areas damaged during construction, including but not limited to, lawn areas, plant beds, trees and shrubs, sidewalks, patios and courtyards. Damaged plant material shall be replaced in kind. Any desire by contractor for pruning, removal of plant material, changes in tree protection, etc., not described in the drawings must be approved by the Engineer prior to any such actions. The Contractor shall be responsible for the actions of his subcontractors with regard to protection of the landscape.
- Contractor shall be responsible for keeping all construction activity within the project limits and staging areas. Any changes in staging areas or site access must have prior approval by the Engineer. Parking is allowed in approved spaces ONLY. No parking is allowed on lawn areas, sidewalks, or courtyards.
- Contractor shall bridge all access and staging areas including but not limited to brick paving, planting beds, grass areas, sidewalks, curbs, etc. Contractor will provide bridging materials, min. 3/4" x 4' x 8' sheet plywood for up to 9,000 lbs. and loads over 9,000 lbs. two layers of 3/4" sheet are required. An inspection of existing conditions will be made prior to installation and documented. It shall be the contractor's responsibility to return all damaged areas to preconstruction conditions at the completion of the project.
- Contractor is responsible for obtaining all permits and inspections and associated fees.
- The contractor shall provide Concrete "Jersey" Barriers when any of the grating is removed for access into the basement. The owner will block access to the alley during any crane operation. The contractor shall barricade the opening all other times when the grating has been removed.
- The contractor shall provide fans to ventilate the welding fumes during construction. Welding will be allowed when the building is occupied.
- Contractor to provide proposed work schedule including numbers of personnel and expected hours of construction for each day of construction. Schedule to be reviewed at pre-construction conference.
- All employees and sub-contractors shall wear a picture identification stating company name, employee name, and company phone number.
- CONTRACTOR TO PROVIDE 24 HOUR CONTACT TO RESPONSIBLE PROJECT MANAGER DURING ENTIRE CONSTRUCTION PERIOD AND THROUGHOUT 1-YEAR WARRANTY PERIOD.

PROJECT SCOPE

THIS PROJECT WILL CONSIST OF REPLACING SIX GAS PACKED UNITS AND ONE SPLIT SYSTEM UNIT. CONNECT NEW UNITS TO EXISTING DUCTWORK AND DIFFUSERS. RTUH WILL RECEIVE NEW DIFFUSERS. CLEAN EXISTING DUCT IN BASE BID.

PROVIDE ALTERNATES AS FOLLOWS:

ALTERNATE #1: PROVIDE VAV OPTION FOR REPLACEMENT OF RTUH. THIS SHALL INCLUDE NEW UNIT, DUCTWORK, TERMINALS AND POWER.

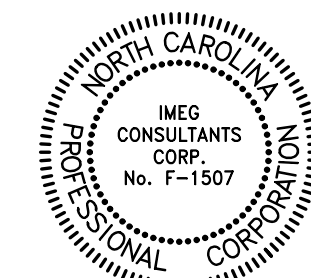
ALTERNATE #2: PROVIDE NEW EXHAUST FANS FOR GROUP TOILETS. CONNECT TO EXISTING DUCTWORK.

ALL WORK TO BE FINISHED BY MAY 21, 2027

PROJECT SCHEDULE

ADVERTISE FOR BIDS: AUGUST 16, 2026
PRE-BID MEETING AT SITE: SEPTEMBER 3, 2026 AT 1:00 PM
BID OPENING: SEPTEMBER 22, 2026 AT 1:00 PM
PROJECT COMPLETION: MAY 15, 2027

PROJECT NO.:
25007326.01



8/13/2026

IMEG
3021 BLUE RIDGE ROAD, SUITE
RALEIGH, NC 27602
PH: (919) 571-1111

HVAC RENOVATION
GREEN ROAD COMMUNITY CENTER GYM
4201 GREEN ROAD
RALEIGH
NORTH CAROLINA

CONTENTS:
COVERSHEET

DATE:
AUGUST 13, 2026

DESIGNER: GWD
ENGINEER: BWF

SHEET NO.

C-1

PACKAGED HEATING/COOLING UNIT SCHEDULE

MARK	BASIS OF DESIGN	FAN CFM	SP	HP	O/A CFM	COOLING CAPACITY TOTAL	COOLING CAPACITY SENSIBLE	GAS HEATING INPUT	GAS HEATING OUTPUT	ELECTRICAL POWER/MCA/MOCP	COOLING EFFICIENCY	HEATING EFFICIENCY	WEIGHT	NOTES
RTU-1	TRANE YHK20	4000	10"	3.0	800	1410 MBH	1117 MBH	150 MBH	1215 MBH	208/3 66 90	15.8 EER	81% EFFIC.	1246 LBS	H-0
RTU-2	TRANE YHK36	1200	10"	0.75	200	436 MBH	33.8 MBH	80 MBH	64.8 MBH	208/3 23 30	16.4 SEER2	81% EFFIC.	767 LBS	I-7.9J
RTU-3	TRANE YHK36	1200	10"	0.75	200	436 MBH	33.8 MBH	80 MBH	64.8 MBH	208/3 23 30	16.4 SEER2	81% EFFIC.	767 LBS	I-7.9J
AC-5	TRANE YHK36	1200	10"	0.75	200	436 MBH	33.8 MBH	80 MBH	64.8 MBH	208/3 23 30	16.4 SEER2	81% EFFIC.	767 LBS	I-7.9
AC-6	TRANE YHK160	6000	10"	Ø 30	850	2020 MBH	1652 MBH	250 MBH	2025 MBH	208/3 79 110	18.0 EER	81% EFFIC.	292 LBS	I-9J2
AC-7	TRANE YHK160	6000	10"	Ø 30	850	2020 MBH	1652 MBH	250 MBH	2025 MBH	208/3 79 110	18.0 EER	81% EFFIC.	292 LBS	I-9J2

NOTES:

1. PROVIDE WITH HEAVY DUTY FUSIBLE DISCONNECT SWITCH.
2. PROVIDE AND HAIL GUARDS.
3. PROVIDE WITH 2" MERV 13 FILTERS.
4. PROVIDE WITH OUTSIDE AIR INTAKE WEATHER HOOD WITH WASHABLE METAL FILTER.
5. PROVIDE WITH BAGNET P INTERFACE. CONTROL BY BAS.
6. PROVIDE WITH GAS REGULATOR.
7. PROVIDE WITH HOT GAS RE-HEAT COIL AND WALL MOUNTED HUMIDISTAT.
8. PROVIDE ENTHALPHY ECONOMIZER WITH BAROMETRIC RELIEF.
9. PROVIDE WITH CARBON DIOXIDE MONITORING SYSTEM CONTROLS FOR DEMAND CONTROL VENTILATION.
10. PROVIDE WITH CONTROLS FOR VAV APPLICATION.
11. PROVIDE WITH ROOF CURB.
12. PROVIDE WITH ACCESSORIES FOR SINGLE ZONE VAV.

SPLIT-SYSTEM HEAT PUMP SCHEDULE

INSIDE UNIT							OUTSIDE UNIT												
MARK	BASIS OF DESIGN	FAN		SUPP. HEAT	ELECTRICAL			MARK	BASIS OF DESIGN	CAPACITY		HEATING CAPACITY	ELECTRICAL			EFFICIENCY		NOTES	
		CFM	SP/HP		POWER	FLA	MOCP			TOTAL	SENSIBLE		POWER	FLA	MOCP	COOLING	HEATING		
AHU-8	TRANE STED07AC3	950	0.5	1	7.2 KW	208/3	32.0	35	HP-8	TRANE STWA4060	571 MBH	425 MBH	34.8 MBH	208/3	22.8	35	14.3 SEER2	8.0 HSPF2	I-6

NOTES:

1. PROVIDE WITH FUSIBLE DISCONNECT ON INDOOR AND OUTDOOR UNITS.
2. PROVIDE WITH SINGLE POINT ELECTRICAL CONNECTION.
3. CONTROL BY BAS.
4. SET OUTSIDE AIR TO 230 CFM.
5. ROUTE CONDENSATE TO EXTERIOR SPLASH BLOCK.
6. PROVIDE WITH LOW AMBIENT CONTROLS FOR OPERATION DOWN TO 0 DEGREES FAHRENHEIT.

VAV SCHEDULE

MARK	BASIS OF DESIGN	COOLING AIRFLOW	HEATING AIRFLOW	MINIMUM AIRFLOW	ELECTRIC HEAT	INLET	OUTLET	VOLTAGE	PHASE	NOTES
VAV-11	TRANE VCE-4	1200	960	180	7.5 KW	14	14	208	3	I-4
VAV-12	TRANE VCE-10	375	260	50	2.0 KW	10	10	208	1	I-4
VAV-13	TRANE VCE-2	700	660	125	5.25 KW	12	12	208	3	I-4
VAV-14	TRANE VCE-10	400	280	55	2.25 KW	10	10	208	1	I-4
VAV-15	TRANE VCE-10	325	320	60	2.5 KW	10	10	208	1	I-4
VAV-16	TRANE VCE-14	1000	720	135	5.75 KW	14	14	208	3	I-4

NOTES:

1. PROVIDE WITH 120 VOLT TO 24 VOLT TRANSFORMER WITH DISCONNECT.
2. PROVIDE WITH FIELD MOUNTED CONTROLS.
3. CONTROL VIA BUILDING AUTOMATION SYSTEM.
4. PROVIDE WITH INLET AIR FLOW PROBE.
5. ELECTRIC STRIP MORE THAN 25 KW SHALL BE TWO STAGE. 25 KW AND LESS SHALL BE ONE STAGE.

GRILLE & DIFFUSER SCHEDULE

MARK	BASIS OF DESIGN	SERVICE	TYPE	MAX. CFM	FACE SIZE	NECK SIZE	NOTES
A	PRICE SPD	SUPPLY	PLAQUE LAY-IN	450	24X24	12"	I-2
B	PRICE SPD	SUPPLY	PLAQUE LAY-IN	300	24X24	10"	I-2
C	PRICE SPD	SUPPLY	PLAQUE LAY-IN	200	24X24	8"	I-2
D	PRICE SPD	SUPPLY	PLAQUE LAY-IN	100	24X24	6"	I-2
E	PRICE SPD	SUPPLY	DUCT MOUNTED	100	10X4	10"	I-2.4
RA	PRICE SPD	RETURN	LOUVERED LAY-IN	1200	24X24	SEE DWG	I-2

NOTES:

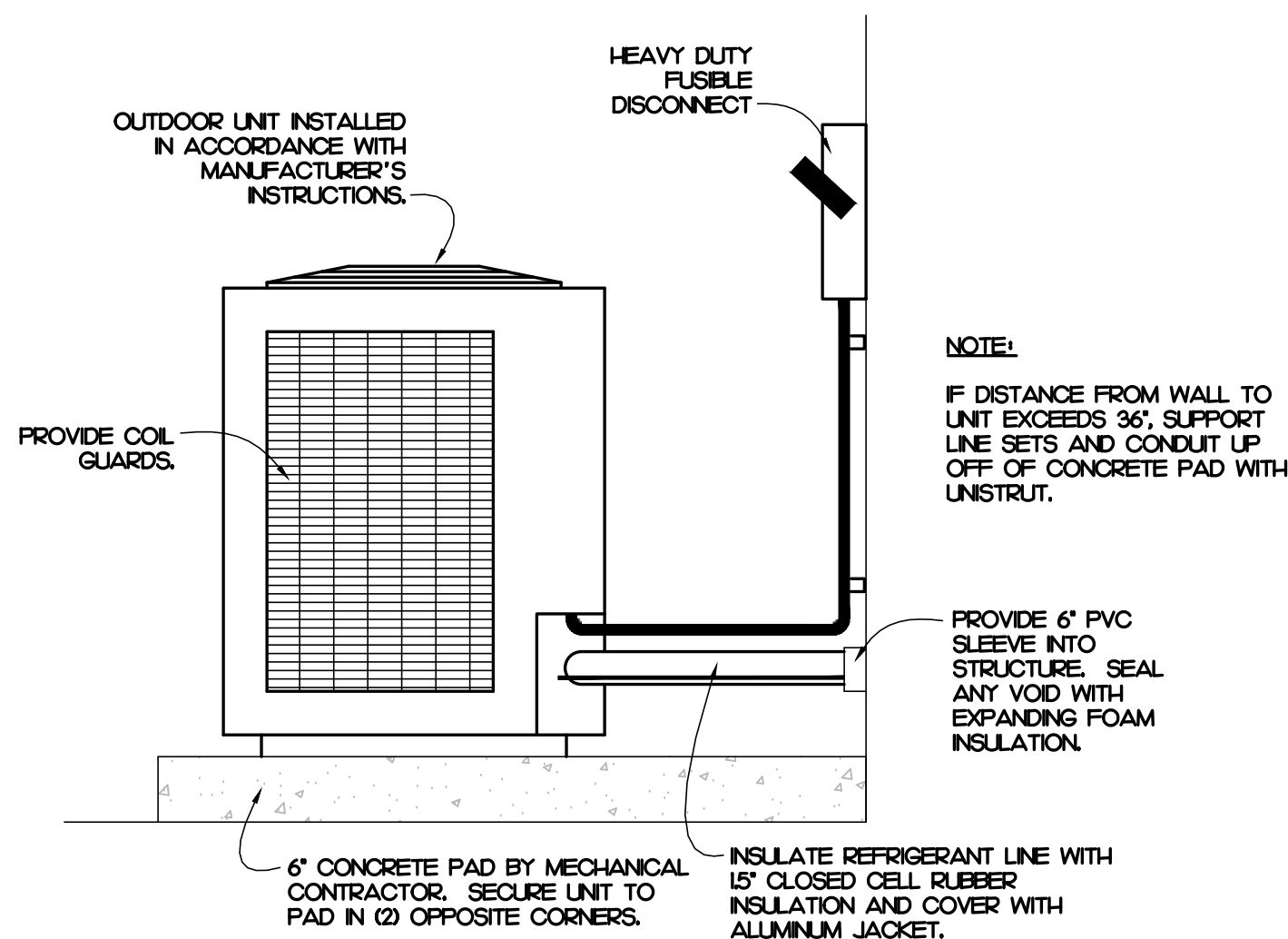
1. PROVIDE WHITE FINISH.
2. PROVIDE WITH INSULATED SHEET METAL PLENUM.
3. PROVIDE WITH FRAME FOR DUCT MOUNTING.
4. PROVIDE WITH OPPOSED BLADE DAMPER.

EXHAUST FAN SCHEDULE

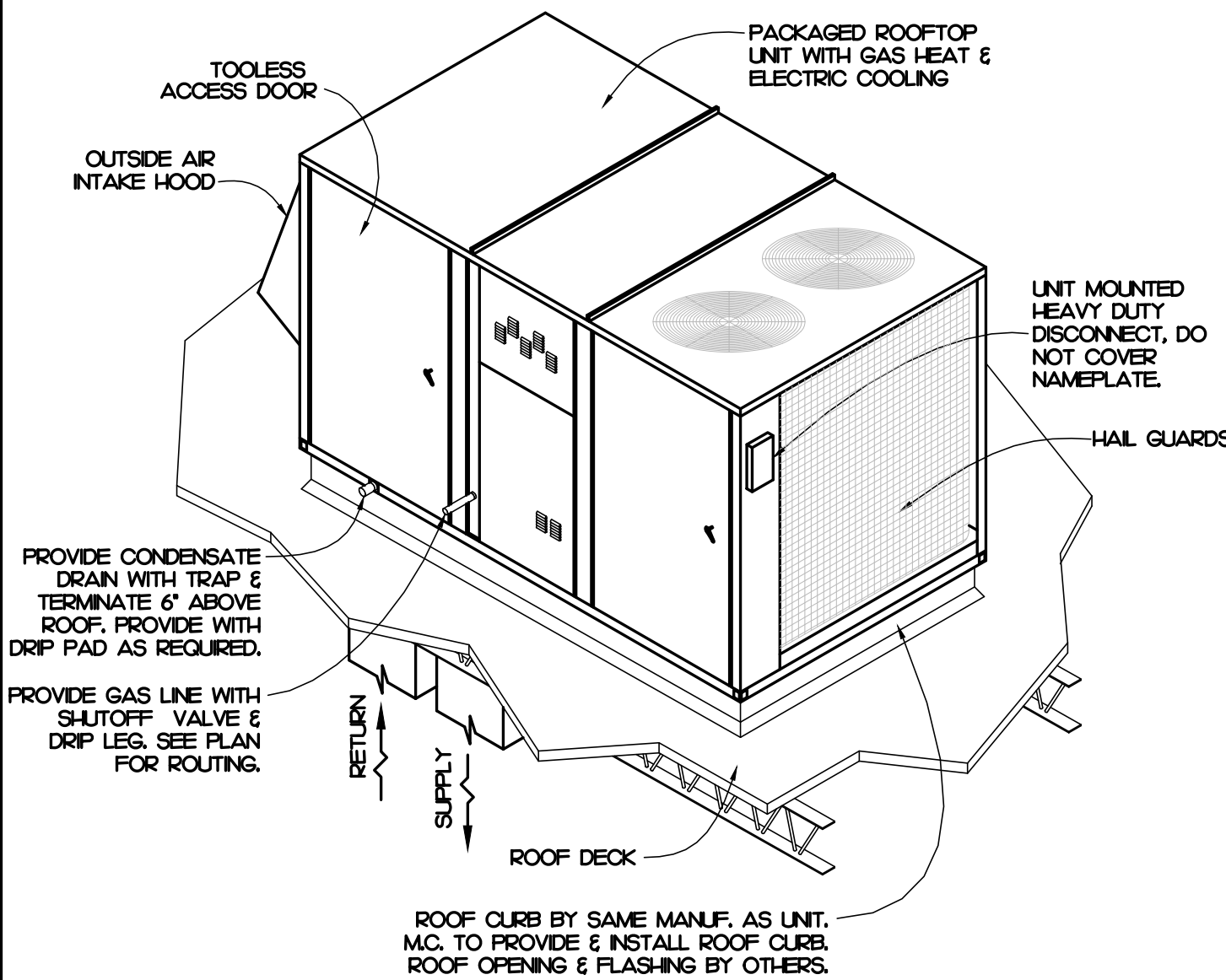
MARK	BASIS OF DESIGN	SERVICE	TYPE	CFM	RPM	HP	SP.	POWER	NOTES
EF-1	PRICE SPD	RESTROOM	CABINET FAN	200	1500	182 Watts	0.25"	120/1	I-3

NOTES:

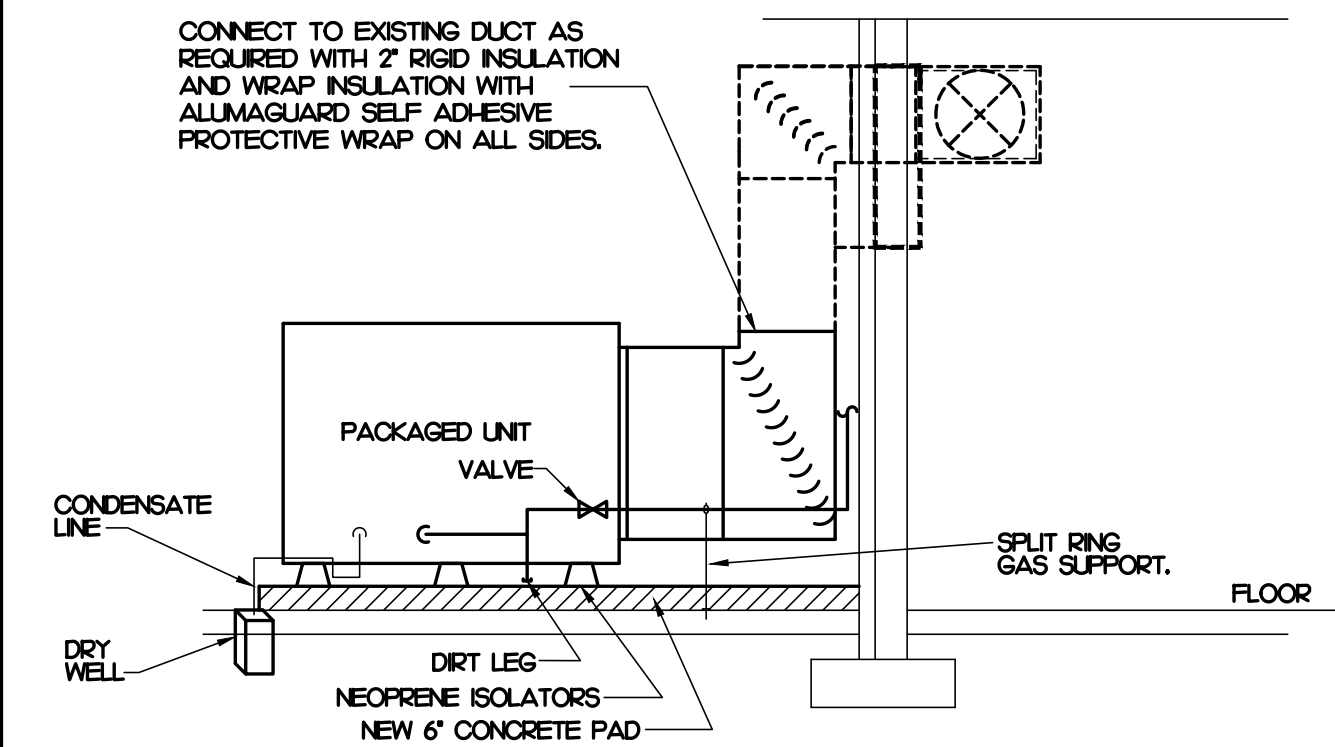
1. PROVIDE WITH DISCONNECT SWITCH.
2. PROVIDE WITH BACKDRAFT DAMPER.
3. CONTROL VIA BAS.



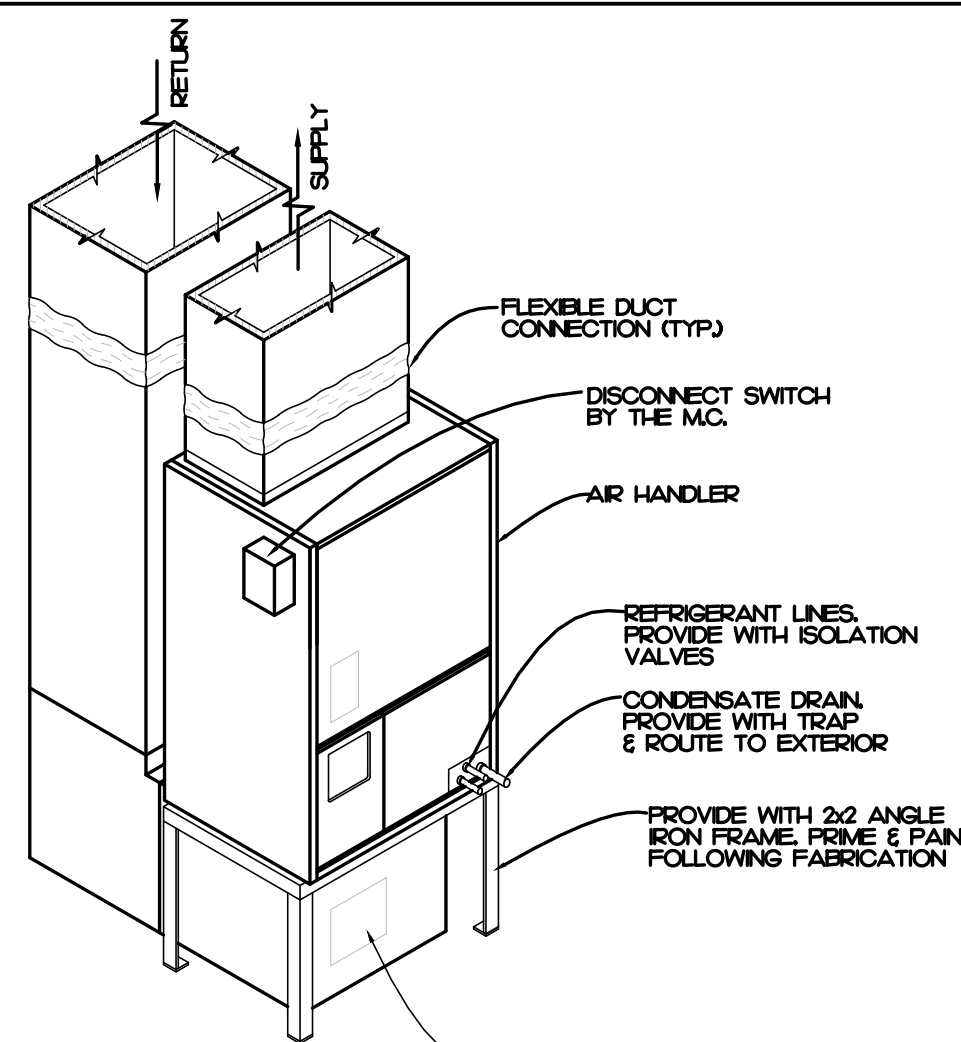
HEAT PUMP
01 - OUTDOOR UNIT DETAIL
SCALE: N.T.S.



02 - ROOF GAS PACK DETAIL
SCALE: N.T.S.



03 - GROUND MOUNTED AC UNIT DETAIL
SCALE: N.T.S.



HEAT PUMP
04 - AHU DETAIL
SCALE: N.T.S.

GENERAL NOTES

1. ALL WORK SHALL BE DONE IN ACCORDANCE WITH THE LATEST EDITION OF THE STATE CODE, ALL LOCAL AND OTHER APPLICABLE CODES.
2. ANY PERMITS AND INSPECTION FEES SHALL BE SECURED AND PAID FOR BY THE MECHANICAL CONTRACTOR (MC).
3. ALL WORK SHALL BE PERFORMED BY EXPERIENCED AND SKILLED CRAFTSMAN. THE MC SHALL COORDINATE ALL OF HIS WORK WITH ALL OTHER CONTRACTORS.
4. THE MECHANICAL PLANS AND SPECIFICATIONS SHALL BE THOROUGHLY REVIEWED PRIOR TO PURCHASING MATERIALS AND INSTALLATION. ALL DISCREPANCIES OR INTERFERENCES SHALL BE BROUGHT TO THE ENGINEERS' ATTENTION.
5. THESE PLANS ARE DIAGRAMMATIC AND MAY NOT SHOW MINOR DETAILS AND LOCATIONS. FOR DIMENSIONS, REFER TO THE ARCHITECTURAL PLANS.
6. THE MC SHALL BE RESPONSIBLE FOR ALL ELECTRICAL STARTERS, INTERLOCKS, CONTROL WIRING. THE ELECTRICAL CONTRACTOR SHALL PROVIDE POWER WIRING, CONDUIT FROM THE DISCONNECT TO MC EQUIPMENT. THE MC SHALL BE RESPONSIBLE FOR ALL FINAL CONNECTION TO HIS EQUIPMENT.
7. INSTALL FLEXIBLE CONNECTORS ON SUPPLY AND RETURN DUCTWORK AT ALL AIR HANDLING UNITS.
8. INSTALL TURNING VANES IN SUPPLY DUCTS AT ELBOWS. PROVIDE BALANCING AND SPLITTER DAMPERS WHERE SHOWN AND AS REQUIRED FOR SYSTEM BALANCING.
9. ALL THERMOSTATS, WIRING AND CONDUIT ARE TO BE FURNISHED BY THE MC. MOUNT THERMOSTATS 4'-0" ABOVE THE FLOOR, UNLESS OTHERWISE NOTED.
10. THE MC SHALL INSURE THAT ALL MECHANICAL EQUIPMENT INSTALLED UNDER HIS CONTRACT SHALL OPERATE FREE OF OBJECTIONABLE NOISE AND VIBRATION.
11. THE MC SHALL KEEP THE PREMISES CLEAR OF DEBRIS FROM HIS WORK DURING CONSTRUCTION AND LEAVE THE AREA AND BUILDING CLEAN AT THE COMPLETION OF HIS WORK. HE SHALL ALSO LEAVE CLEAN ALL EXPOSED EQUIPMENT IN HIS CONTRACT.
12. ALL DUCTWORK SIZES SHOWN ARE ACTUAL SHEET METAL DIMENSIONS. EXTERNALLY WRAP ALL DUCT WITH 3" FOL-BACKED INSULATION FOR A MINIMUM OF R-8.
13. ALL GAS PIPING SHALL BE SCHEDULE 40 BLACK STEEL INSTALLED IN ACCORDANCE WITH ALL CODES. THE MC SHALL COORDINATE GAS PIPE CONNECTION SIZE WITH EQUIPMENT.
14. MECHANICAL CONTRACTOR SHALL WORK WITH TEST AND BALANCE CONTRACTOR TO REMEDY ANY DIFFERENCES TO INCLUDE FAN DRIVE CHANGES, INSTALLATION OF DAMPERS OR OTHER MINOR DUCT MODIFICATIONS TO PROVIDE AIRFLOW TO WITHIN +/- 10% OF THE DESIGN VALUES LISTED ON THESE PLANS.
15. THE AIR HANDLING UNIT SHALL OPERATE AT ALL TIMES DURING OCCUPIED HOURS.
16. THE MECHANICAL CONTRACTOR SHALL PROVIDE THE ENGINEER WITH A SET OF AS-BUILT DRAWINGS UPON COMPLETION OF JOB.
17. THE MECHANICAL CONTRACTOR SHALL PROVIDE THE ENGINEER WITH A SET OF DUCT SHOP DRAWINGS FOR APPROVAL.
18. THE MECHANICAL CONTRACTOR SHALL PROVIDE THE ENGINEER WITH A BALANCE REPORT BY A CERTIFIED TEST AND BALANCE COMPANY.
19. PROVIDE PERMIT LABEL ENGRAVED PLASTIC LAMINATE MECHANICALLY FASTENED TO OUTDOOR UNITS.
20. LABEL CEILING GRID WHERE EQUIPMENT IS LOCATED ABOVE LAY-IN CEILING WITH EQUIPMENT IDENTIFIER. ALSO LABEL ALL TEMPERATURE SENSORS AND THERMOSTATS WITH EQUIPMENT IDENTIFIER.

SYMBOL LEGEND

	NEW SHEET METAL DUCT
	EXISTING SHEET METAL DUCT
	NEW FLEX DUCT
	TEMPERATURE SENSOR - MOUNTED 48" ABOVE FINISHED FLOOR
	EXISTING GAS PIPING
	EXISTING DUCT MOUNTED SMOKE DETECTOR

OUTSIDE AIR SUMMARY

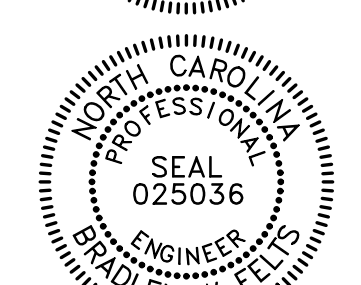
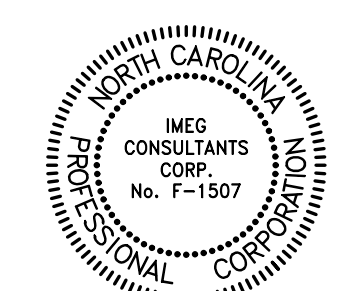
- REQUIRED:
- MEETING ROOM: 675 SQFT • 0.06 CFM/SQFT • 40 PERSONS • 5 CFM/PERSON • 190 CFM
- FITNESS: 675 SQFT • 0.06 CFM/SQFT • 6 PERSONS • 20 CFM/PERSON • 160 CFM
- OFFICE: 837 SQFT • 0.06 CFM/SQFT • 4 PERSONS • 5 CFM/PERSON • 40 CFM
- CIRCULATION/TOILETS/STORAGE: 340 SQFT X 0.12 CFM/SQFT • 409 CFM
- PROVIDED:
- RTU 1: 800 CFM (NOTE 1)
- REQUIRED: MEETING ROOM: 1280 SQFT • 0.06 CFM/SQFT • 64 PERSONS • 5 CFM/PERSON • 397 CFM
- PROVIDED: RTU 2 AND 3 AT 200 CFM EACH (NOTE 1)
- REQUIRED: OLD GYM: 769 SQFT • 0.06 CFM/SQFT • 250 PERSONS • 5 CFM/PERSON • 1707 CFM
- PROVIDED: AC-4 AT 1700 CFM (NOTE 1)
- REQUIRED: MEETING ROOM: 540 SQFT • 0.06 CFM/SQFT • 25 PERSONS • 5 CFM/PERSON • 157 CFM
- PROVIDED: AC-5 AT 200 CFM (NOTE 1)
- REQUIRED: NEW GYM: 7341 SQFT • 0.06 CFM/SQFT • 250 PERSON • 5 CFM/PERSON • 16905 CFM
- PROVIDED: AC6 AND AC7 AT 850 CFM EACH (NOTE 1)
- REQUIRED: CORRIDORS/LOCKERS/STORAGE: 1889 SQFT X 0.12 CFM/SQFT • 227 CFM
- PROVIDED: AHU-8 • 230 CFM

TOTAL REQUIRED: 2295 CFM

NOTE 1: UNITS PROVIDED WITH DEMAND CONTROL VENTILATION SEQUENCE TO REDUCE OUTSIDE BY 50% OF REQUIRED VOLUME. CARBON DIOXIDE SENSORS TO BE USED IN CONJUNCTION WITH OUTSIDE AIR ECONOMIZER TO INCREASE OUTDOOR AIRFLOW PAST MINIMUM UP CARBON DIOXIDE LEVEL ABOVE 1000 PPM.

REVISIONS					

PROJECT NO.:
25007326.01



8/13/2026 13

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PH: (919) 571-1111

HVAC RENOVATION
GREEN ROAD COMMUNITY CENTER GYM
4201 GREEN ROAD
RALEIGH
NORTH CAROLINA

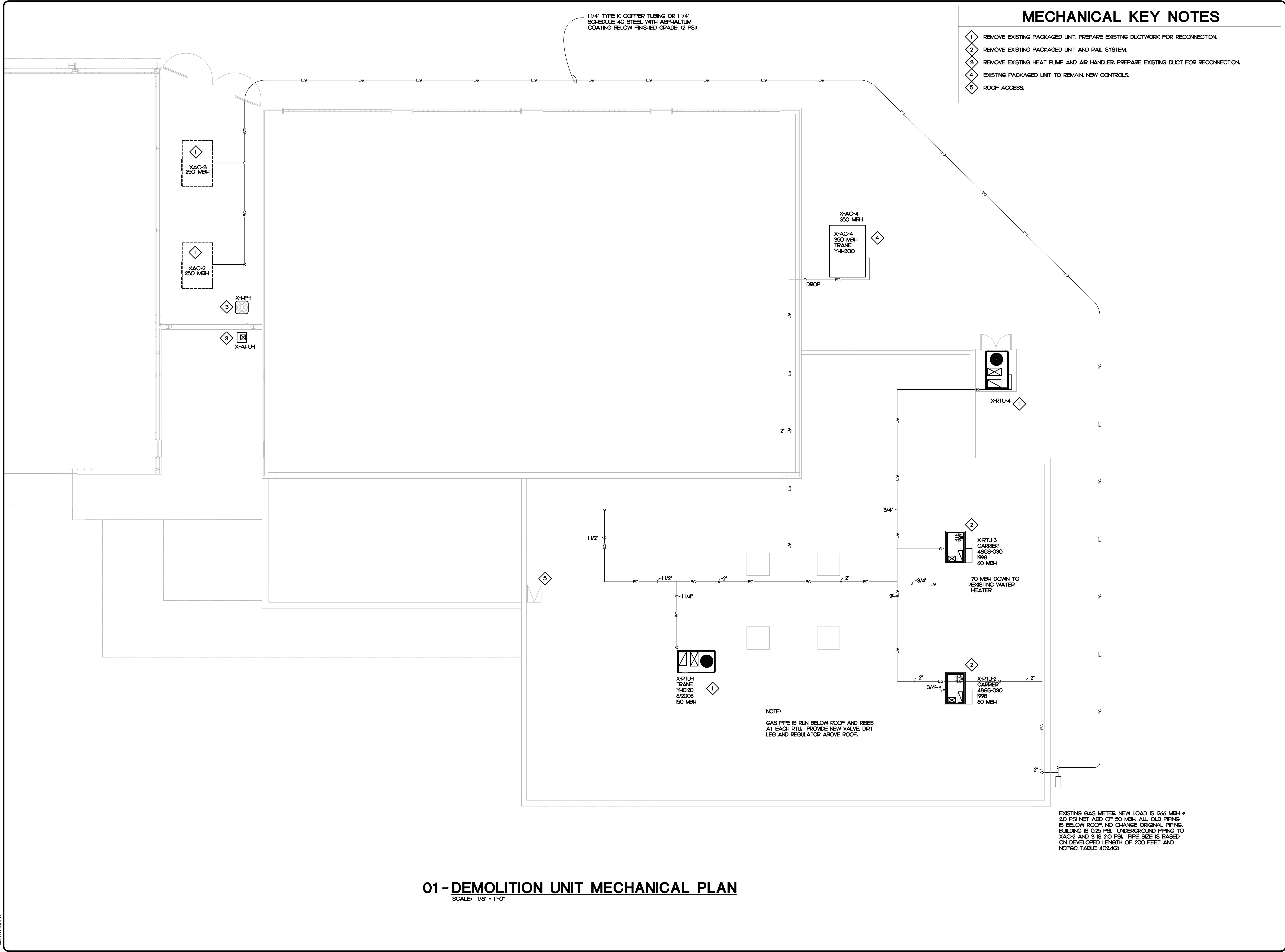
CONTENTS:
MECHANICAL NOTES,
LEGEND, SCHEDULES
AND DETAILS

DATE:
AUGUST 13, 2026

DESIGNER: GWD
ENGINEER: BWF

SHEET NO.

M-1



REVISIONS	

PROJECT NO.:
25007326.01

MECHANICAL
CONSULTANTS
CORP.
No. F-1507

PROFESSIONAL CORPORATION

MECHANICAL
CONSULTANTS
CORP.
No. F-1507

PROFESSIONAL CORPORATION

SEAL
025036

ENGINEER
BRADLEY W. ELLIS

8/13/2026

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HVAC RENOVATION
GREEN ROAD COMMUNITY CENTER GYM
4201 GREEN ROAD
RALEIGH NORTH CAROLINA

CONTENTS:
MECHANICAL
DEMOLITION PLAN

DATE:
AUGUST 13, 2026

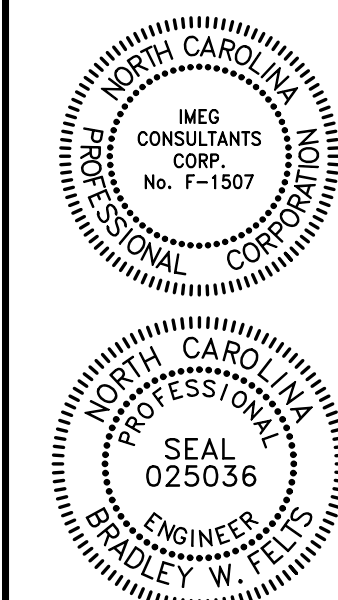
DESIGNER: GWD
ENGINEER: BWF

SHEET NO.
M-2



REMOVE AND REPLACE EXISTING CABINET FAN IN GROUP TOILET.

- PROJECT NO.:
25007326.01



8/13/2026 



HVAC RENOVATION
GREEN ROAD COMMUNITY CENTER GYM
4201 GREEN ROAD
EIGH NORTH CAROL

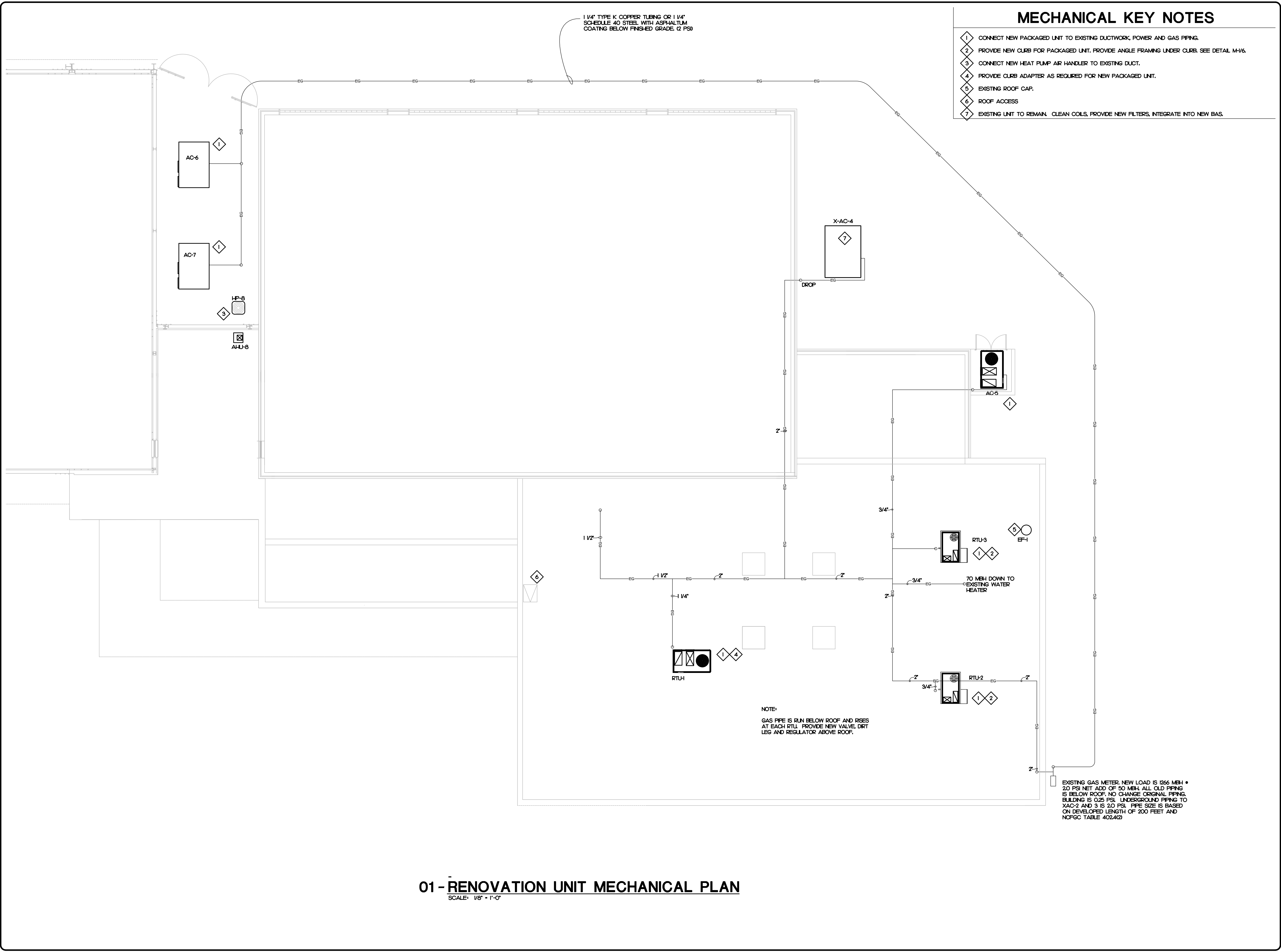
CONTENTS:
BASE BID MECHANICAL
RENOVATION PLAN

DATE:
AUGUST 13, 2026

DESIGNER: **GWD**
ENGINEER: **BWF**

SHEET NO.

M-3



MECHANICAL KEY NOTES

- 1. CONNECT NEW PACKAGED UNIT TO EXISTING DUCTWORK, POWER AND GAS PIPING.
- 2. PROVIDE NEW CURB FOR PACKAGED UNIT. PROVIDE ANGLE FRAMING UNDER CURB. SEE DETAIL M-4/6.
- 3. CONNECT NEW HEAT PUMP AIR HANDLER TO EXISTING DUCT.
- 4. PROVIDE CURB ADAPTER AS REQUIRED FOR NEW PACKAGED UNIT.
- 5. EXISTING ROOF CAP.
- 6. ROOF ACCESS
- 7. EXISTING UNIT TO REMAIN. CLEAN COILS, PROVIDE NEW FILTERS, INTEGRATE INTO NEW BAS.

REVISIONS					

PROJECT NO.:
25007326.01

Professional Seal

Professional Seal

MEG CONSULTANTS CORP. No. F-1507

SEAL 025036

8/13/2026

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HVAC RENOVATION
GREEN ROAD COMMUNITY CENTER GYM
4201 GREEN ROAD
RALEIGH NORTH CAROLINA

CONTENTS:
MECHANICAL
RENOVATION PLAN

DATE:
AUGUST 13, 2026

DESIGNER: GWD
ENGINEER: BWF

SHEET NO.
M-4

MECHANICAL KEY NOTES

- 1

NEW GRILLE PRICED IN THE BASE BID.
- 2

NEW TEMPERATURE SENSOR FOR VAV TERMINAL. SURFACE MOUNT WIRE MOLD ON EXISTING BLOCK WALLS. FISH WIRE IN WALL WHERE SHEET ROCK.
- 3

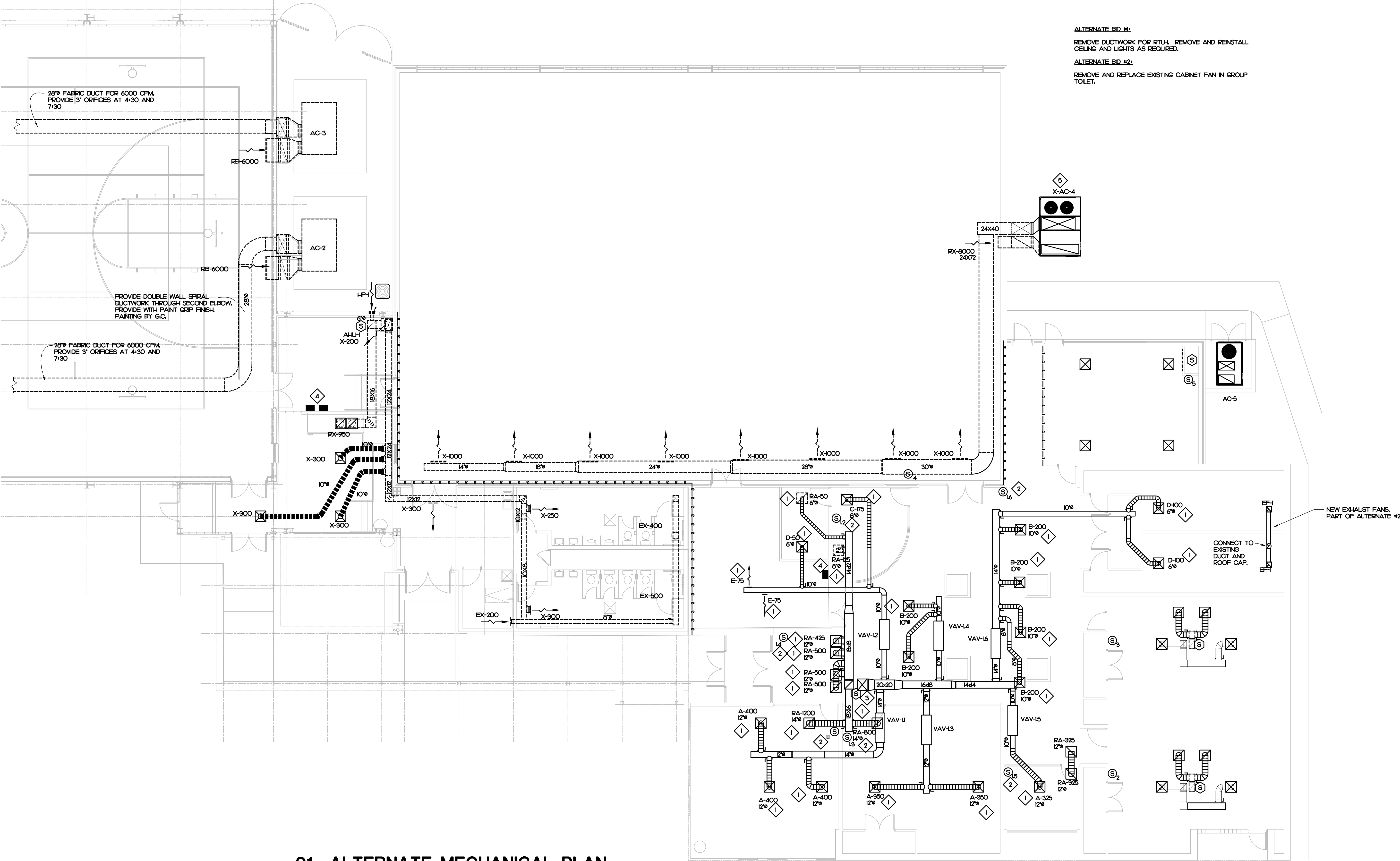
REINSTALL EXISTING DUCT DETECTOR INTO NEW RETURN DROP.
- 4

EXISTING BUILDING AUTOMATION PANEL LOCATIONS.
- 5

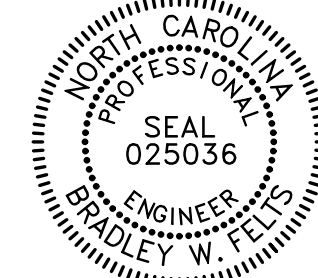
EXISTING 30-TON PACKAGED UNIT TO REMAIN. CLEAN COILS, PROVIDE NEW FILTERS AND INTEGRATE INTO NEW BAS.

ALTERNATE BID #1:
REMOVE DUCTWORK FOR RTU-1. REMOVE AND REINSTALL CEILING AND LIGHTS AS REQUIRED.

ALTERNATE BID #2:
REMOVE AND REPLACE EXISTING CABINET FAN IN GROUP TOILET.



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HVAC RENOVATION
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RALEIGH NORTH CAROLINA

CONTENTS:
ALTERNATE BID
MECHANICAL
RENOVATION PLAN

DATE:
AUGUST 13, 2026

DESIGNER: GWD
ENGINEER: BWF

SHEET NO.

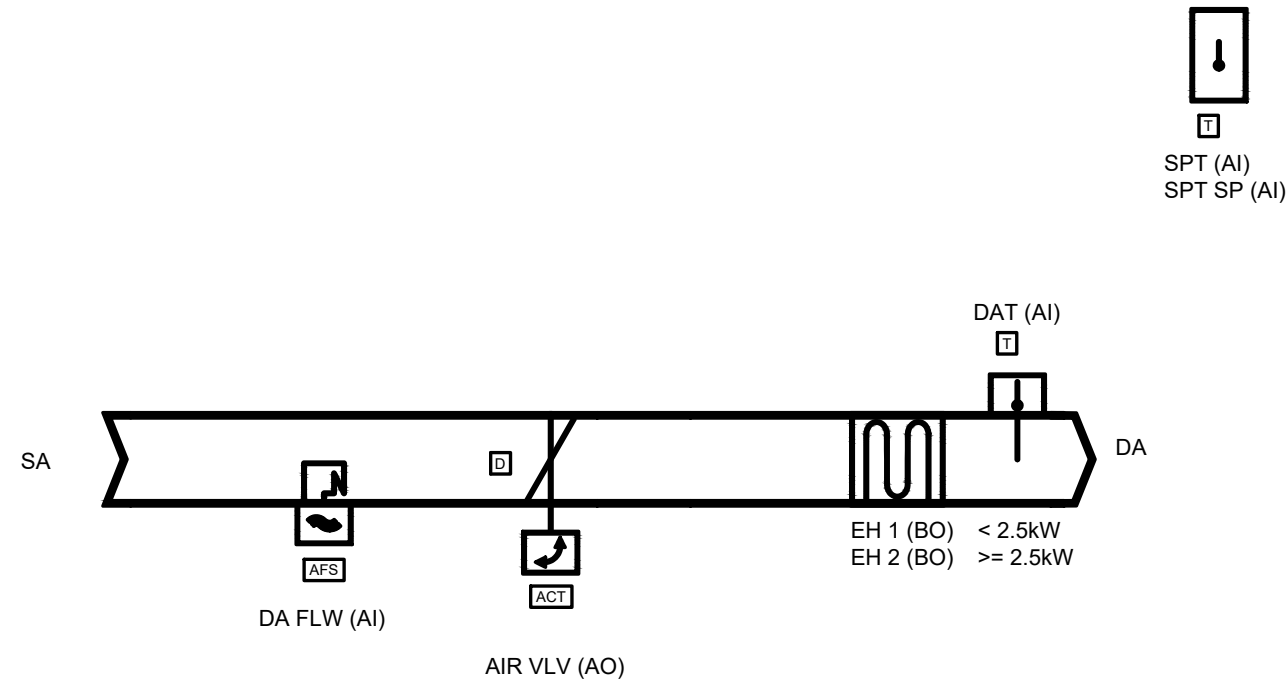
M-5

SEQUENCE OF OPERATIONS:

THE VAV TERMINAL WILL BE SUPPLIED WITH A FIELD MOUNTED BACNET IP CONTROLLER AND DAI5Y CHAINED TO OTHER VAV TERMINALS.

TERMINALS SHALL MODULATE DAMPER AND STAGE ELECTRIC HEAT TO MAINTAIN SPACE TEMPERATURE.

CONTROLS SHALL MONITOR SPACE TEMPERATURE, DISCHARGE TEMPERATURE AND INLET AIRFLOW FOR REPORTING TO BAS.



03 - VAV TERMINAL SCHEMATIC
SCALE: N.T.S.

SEQUENCE OF OPERATIONS:

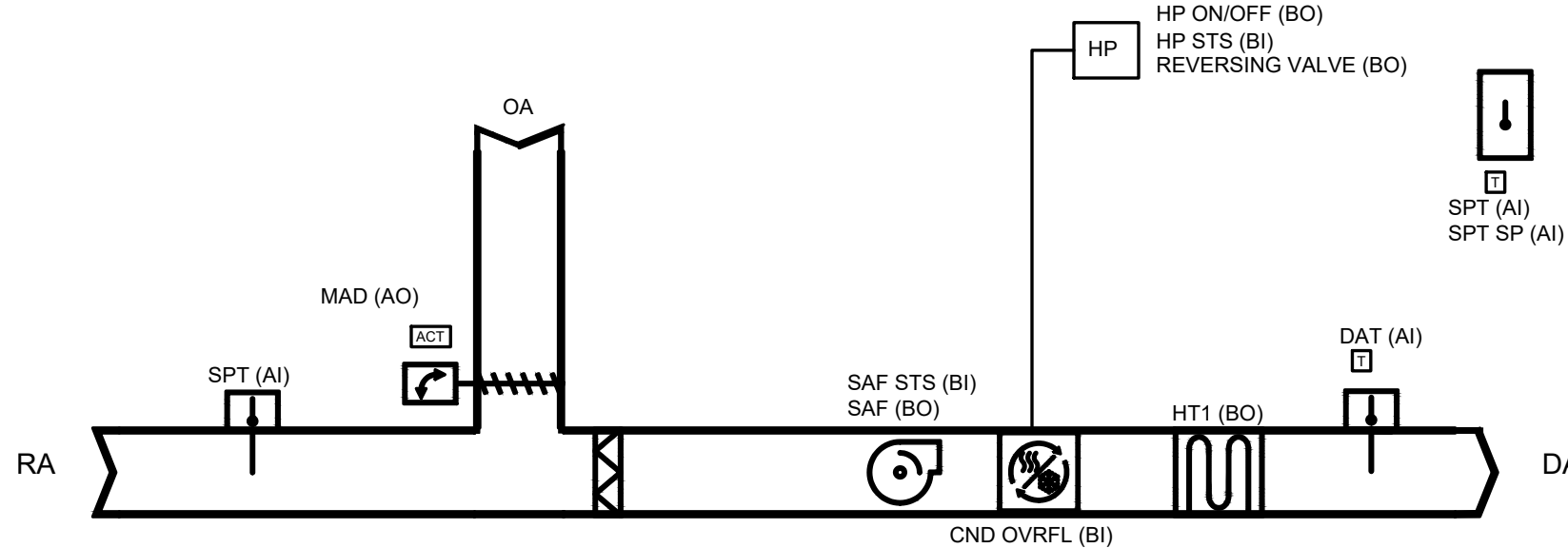
THE SPLIT SYSTEM HEAT PUMP WILL BE CONTROLLED FIELD MOUNTED BACNET IP CONTROLLER.

DURING OCCUPIED HOURS THE UNIT SHALL BE ENABLED AND THE FAN RUN AND OUTSIDE AIR DAMPER OPEN. CONTROLS SHALL INDEX COOLING/HEATING AND ELECTRIC STRIP HEAT TO MAINTAIN SETPOINT.

CONTROLS SHALL MONITOR SPACE TEMPERATURE, DISCHARGE TEMPERATURE FOR REPORTING TO BAS.

CONTROLS SHALL MONITOR ALARM STATE OF SUPPLY FAN, OUTDOOR UNIT AND CONDENSATE FLOAT.

DURING UNOCCUPIED HOURS THE FAN AND OUTDOOR UNIT SHALL CYCLE TO MAINTAIN SET BACK TEMPERATURES. THE OUTSIDE AIR DAMPER SHALL REMAIN CLOSED.



04 - SPLIT SYSTEM CONTROL SCHEMATIC
SCALE: N.T.S.

SEQUENCE OF OPERATIONS:

THE PACKAGED UNITS WILL BE CONTROLLED FACTORY MOUNTED BACNET IP CONTROLLER, EXCEPT AC-4. AC-4 IS EXISTING BACNET MSTP AND SHALL BE INTEGRATED INTO THE NEW CONTROLS. DISCHARGE AND ROOM TEMPERATURE AND HUMIDITY SENSORS SHALL BE WIRED TO THE UNIT CONTROLS.

DURING OCCUPIED HOURS THE UNITS SHALL BE ENABLED AND THE FAN RUN AT LOW SPEED AND OUTSIDE AIR DAMPER OPEN TO MINIMUM POSITION. CONTROLS SHALL INDEX FAN SPEED AND COOLING/HEATING TO MAINTAIN OCCUPIED SETPOINTS.

UNITS SHALL BE PROVIDED IN HOT GAS RE-HEAT COIL TO MAINTAIN HIGH HUMIDITY LIMIT 60% ADJ.

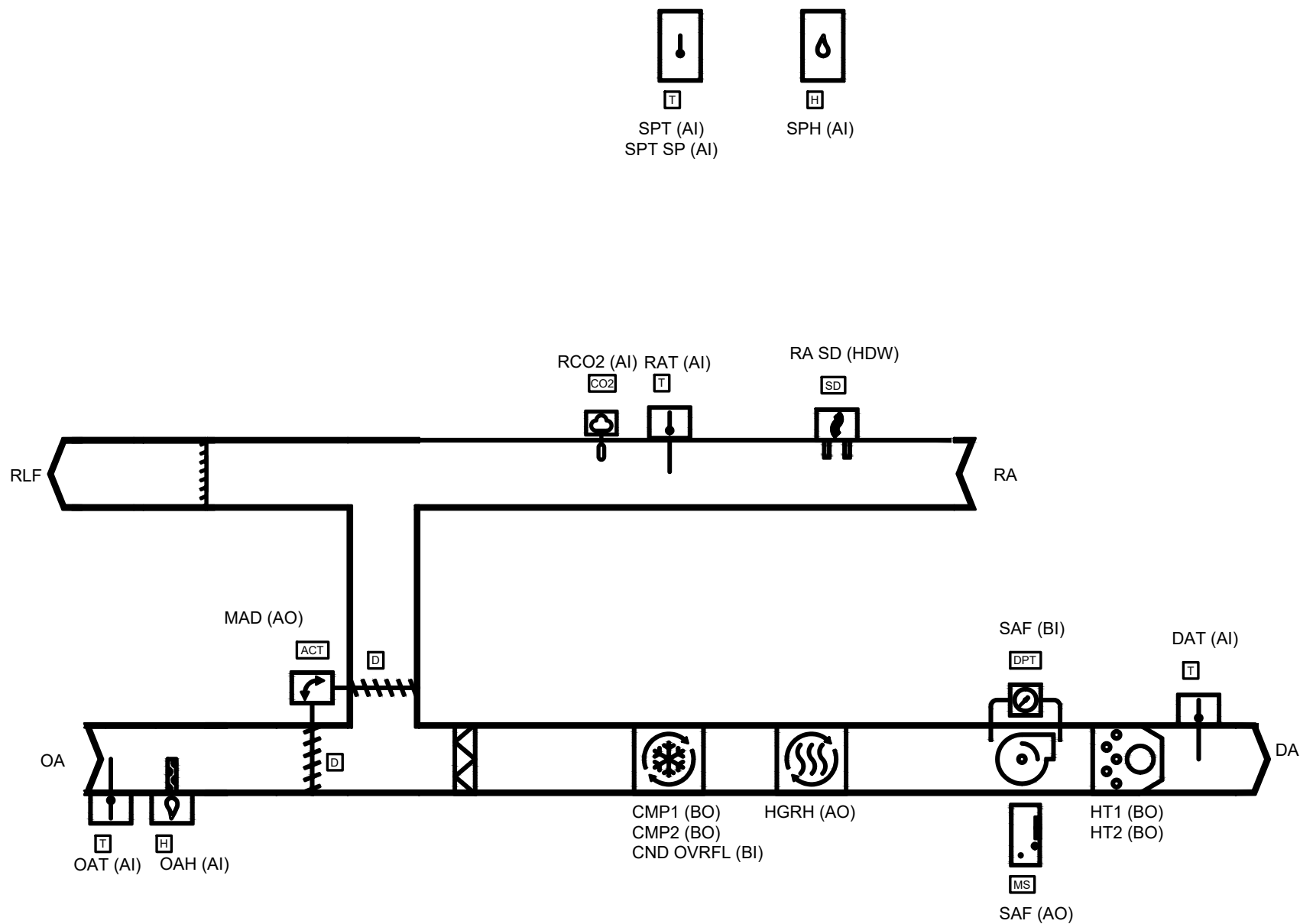
UPON A RISE IN RETURN DUCT CO2 ABOVE 1000 PPM THE OUTSIDE AIR DAMPER SHALL OPEN TILL CO2 LEVEL FALL BELOW 800 PPM.

UNITS SHALL BE PROVIDED WITH ECONOMIZER CONTROL TO PROVIDE FREE COOLING WHEN OUTSIDE CONDITIONS EXIST. REMOTE MOUNTED BAROMETRIC RELIEF DAMPER SHALL RELIEVE EXCESS BUILDING PRESSURE. AC-4 HAS POWERED RELIEF DAMPER.

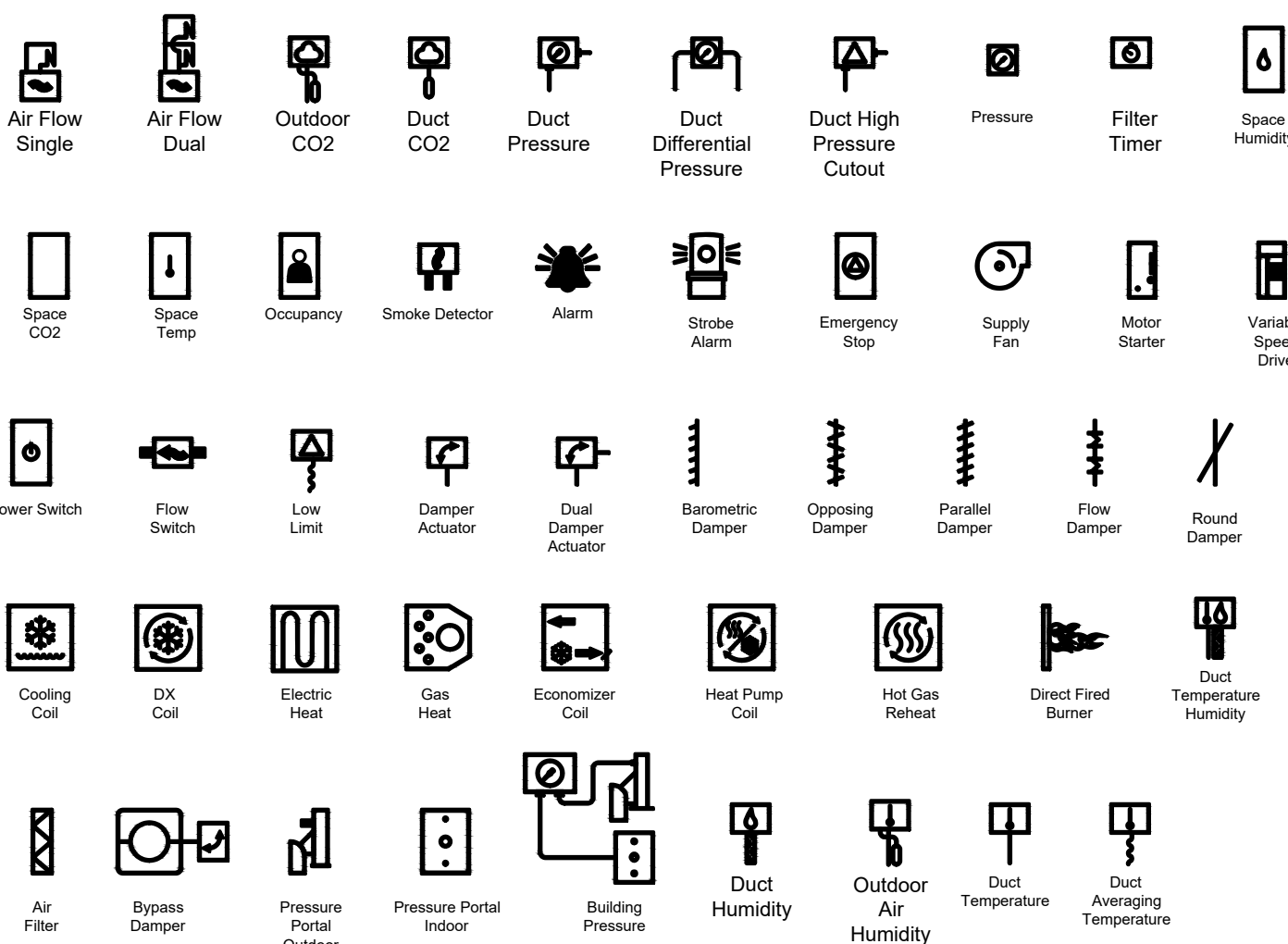
CONTROLS SHALL MONITOR SPACE TEMPERATURE/HUMIDITY, DISCHARGE TEMPERATURE FOR REPORTING TO BAS.

CONTROLS SHALL MONITOR ALARM STATE OF SUPPLY FAN, COMPRESSOR AND CONDENSATE FLOAT.

DURING UNOCCUPIED HOURS THE UNIT SHALL CYCLE TO MAINTAIN SET BACK TEMPERATURES. THE OUTSIDE AIR DAMPER SHALL REMAIN CLOSED.



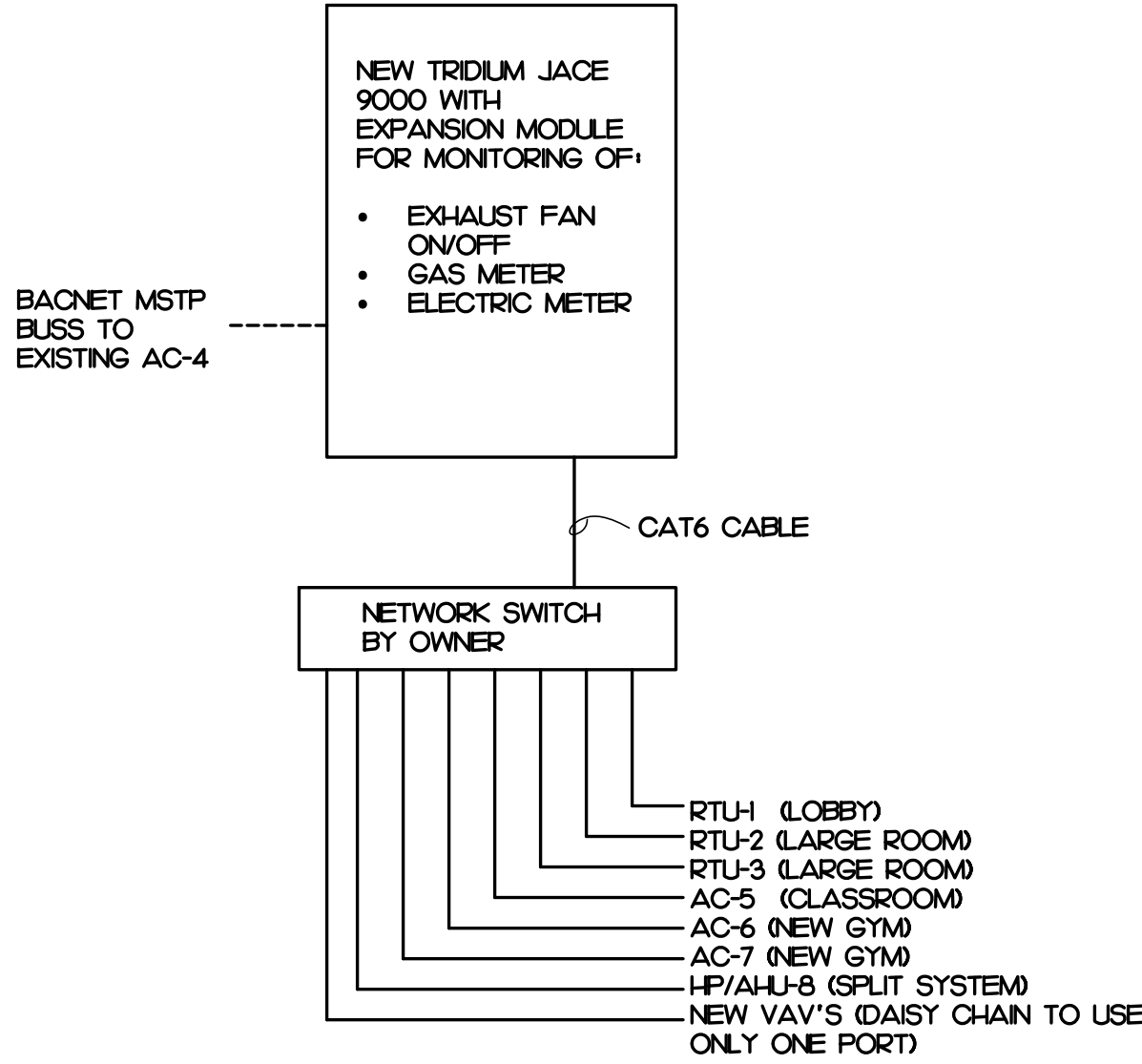
06 - SINGLE ZONE VAV PACKAGED UNIT CONTROL SCHEMATIC
SCALE: N.T.S.



02 - CONTROL LEGEND
SCALE: N.T.S.

CONTROL NOTES:

- ALL WIRING SHALL BE RUN IN CONDUIT.
- MOUNT THERMOSTAT/HUMIDISTAT AND REMOTE SENSORS IN CAGE IN GYMNASIUMS.



01 - BAS ARCHITECTURE
SCALE: N.T.S.

SEQUENCE OF OPERATIONS:

THE PACKAGED UNITS WILL BE CONTROLLED FACTORY MOUNTED BACNET IP CONTROLLER. DISCHARGE AND ROOM TEMPERATURE AND HUMIDITY SENSORS SHALL BE WIRED TO THE UNIT CONTROLS.

DURING OCCUPIED HOURS THE UNITS SHALL BE ENABLED AND THE FAN RUN AND OUTSIDE AIR DAMPER OPEN TO MINIMUM POSITION. CONTROLS SHALL INDEX COOLING/HEATING TO MAINTAIN OCCUPIED SETPOINTS.

UNITS SHALL BE PROVIDED IN HOT GAS RE-HEAT COIL TO MAINTAIN HIGH HUMIDITY LIMIT 60% ADJ.

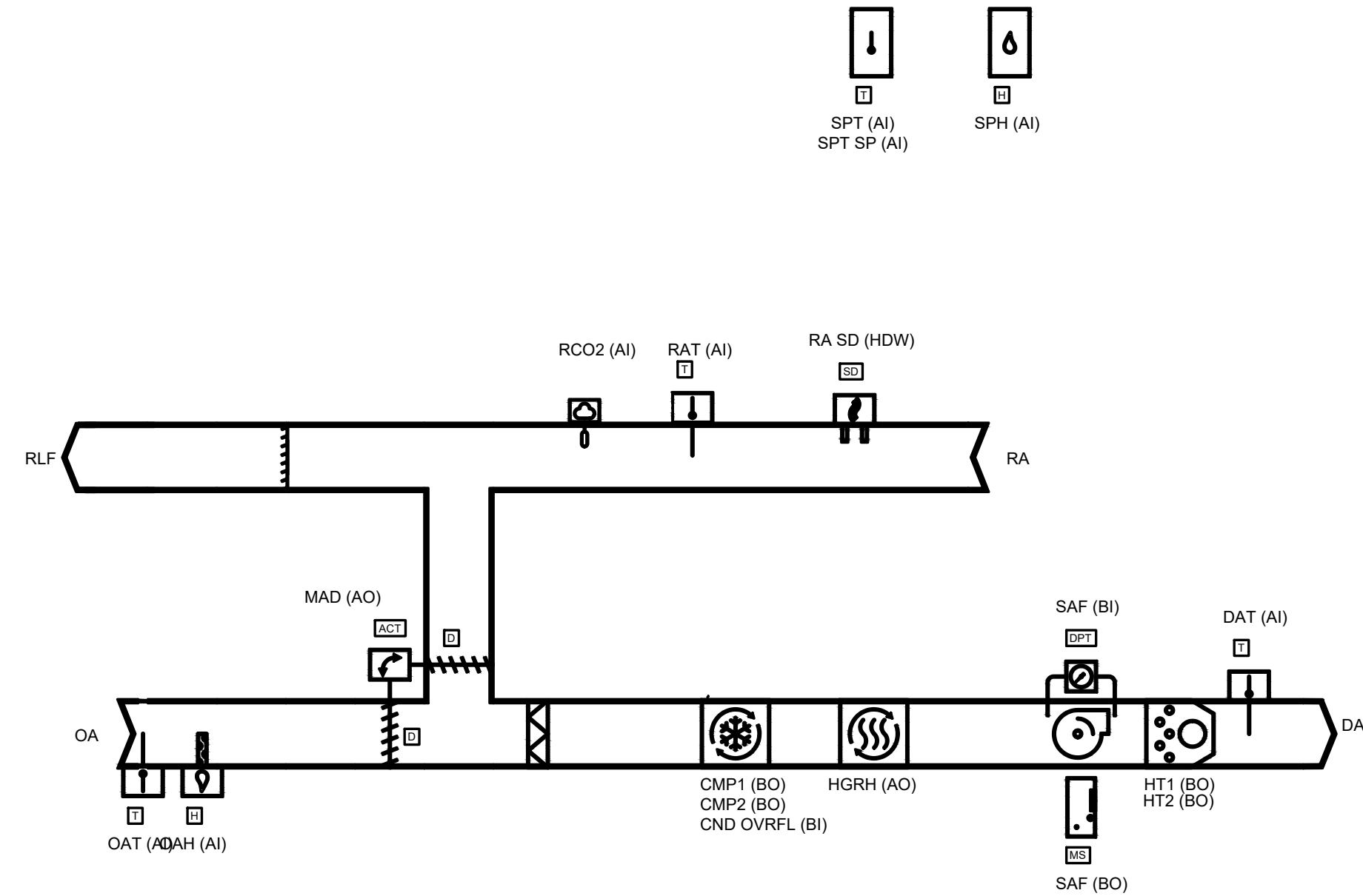
UPON A RISE IN RETURN DUCT CO2 ABOVE 1000 PPM THE OUTSIDE AIR DAMPER SHALL OPEN TILL CO2 LEVEL FALL BELOW 800 PPM.

RTU-1 SHALL BE PROVIDED WITH ECONOMIZER CONTROL TO PROVIDE FREE COOLING WHEN OUTSIDE CONDITIONS EXIST. BAROMETRIC RELIEF DAMPER SHALL CONTROL BUILDING PRESSURE.

CONTROLS SHALL MONITOR SPACE TEMPERATURE, DISCHARGE TEMPERATURE FOR REPORTING TO BAS.

CONTROLS SHALL MONITOR ALARM STATE OF SUPPLY FAN, COMPRESSOR AND CONDENSATE FLOAT.

DURING UNOCCUPIED HOURS THE UNIT SHALL CYCLE TO MAINTAIN SET BACK TEMPERATURES. THE OUTSIDE AIR DAMPER SHALL REMAIN CLOSED.



05 - CONSTANT VOLUME PACKAGED UNIT CONTROL SCHEMATIC
SCALE: N.T.S.

SEQUENCE OF OPERATIONS:

THE PACKAGED UNITS WILL BE CONTROLLED FACTORY MOUNTED BACNET IP CONTROLLER. DISCHARGE TEMPERATURE AND DUCT PRESSURE SENSORS SHALL BE WIRED TO THE UNIT CONTROLS.

DURING OCCUPIED HOURS THE UNITS SHALL BE ENABLED AND THE FAN RUN TO MAINTAIN DUCT PRESSURE. OUTSIDE AIR DAMPER OPEN TO MINIMUM POSITION. COOLING/HEATING TO MAINTAIN DISCHARGE AIR TEMPERATURE.

UNITS SHALL BE PROVIDED IN HOT GAS RE-HEAT COIL TO MAINTAIN HIGH HUMIDITY LIMIT 60% ADJ.

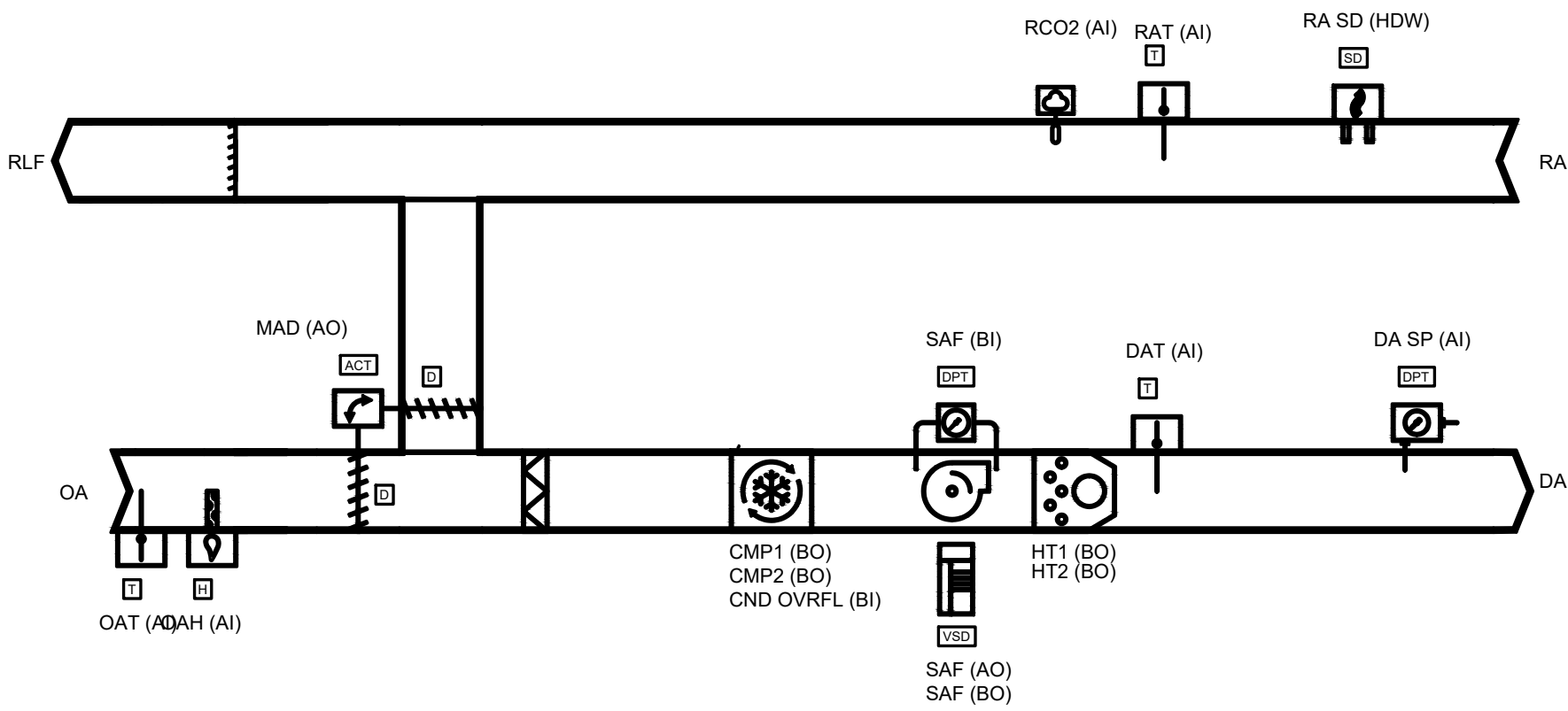
UPON A RISE IN RETURN DUCT CO2 ABOVE 1000 PPM THE OUTSIDE AIR DAMPER SHALL OPEN TILL CO2 LEVEL FALL BELOW 800 PPM.

UNITS SHALL BE PROVIDED WITH ECONOMIZER CONTROL TO PROVIDE FREE COOLING WHEN OUTSIDE CONDITIONS EXIST. A UNIT MOUNTED BAROMETRIC RELIEF DAMPER SHALL RELIEVE EXCESS BUILDING PRESSURE.

CONTROLS SHALL MONITOR SPACE TEMPERATURE/HUMIDITY, DISCHARGE TEMPERATURE FOR REPORTING TO BAS.

CONTROLS SHALL MONITOR ALARM STATE OF SUPPLY FAN, COMPRESSOR AND CONDENSATE FLOAT.

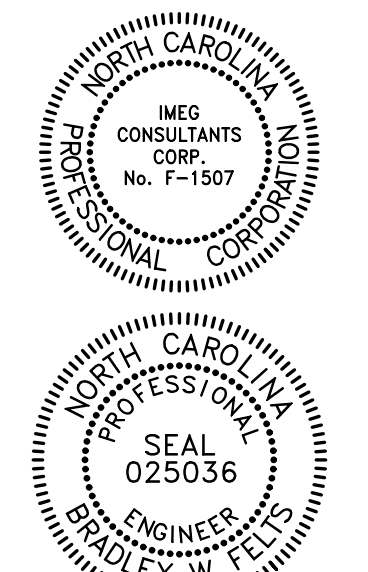
DURING UNOCCUPIED HOURS THE UNIT SHALL CYCLE TO MAINTAIN SET BACK TEMPERATURES. THE OUTSIDE AIR DAMPER SHALL REMAIN CLOSED.



07 - VAV PACKAGED UNIT CONTROL SCHEMATIC
SCALE: N.T.S.

REVISIONS

PROJECT NO.:
25007326.01



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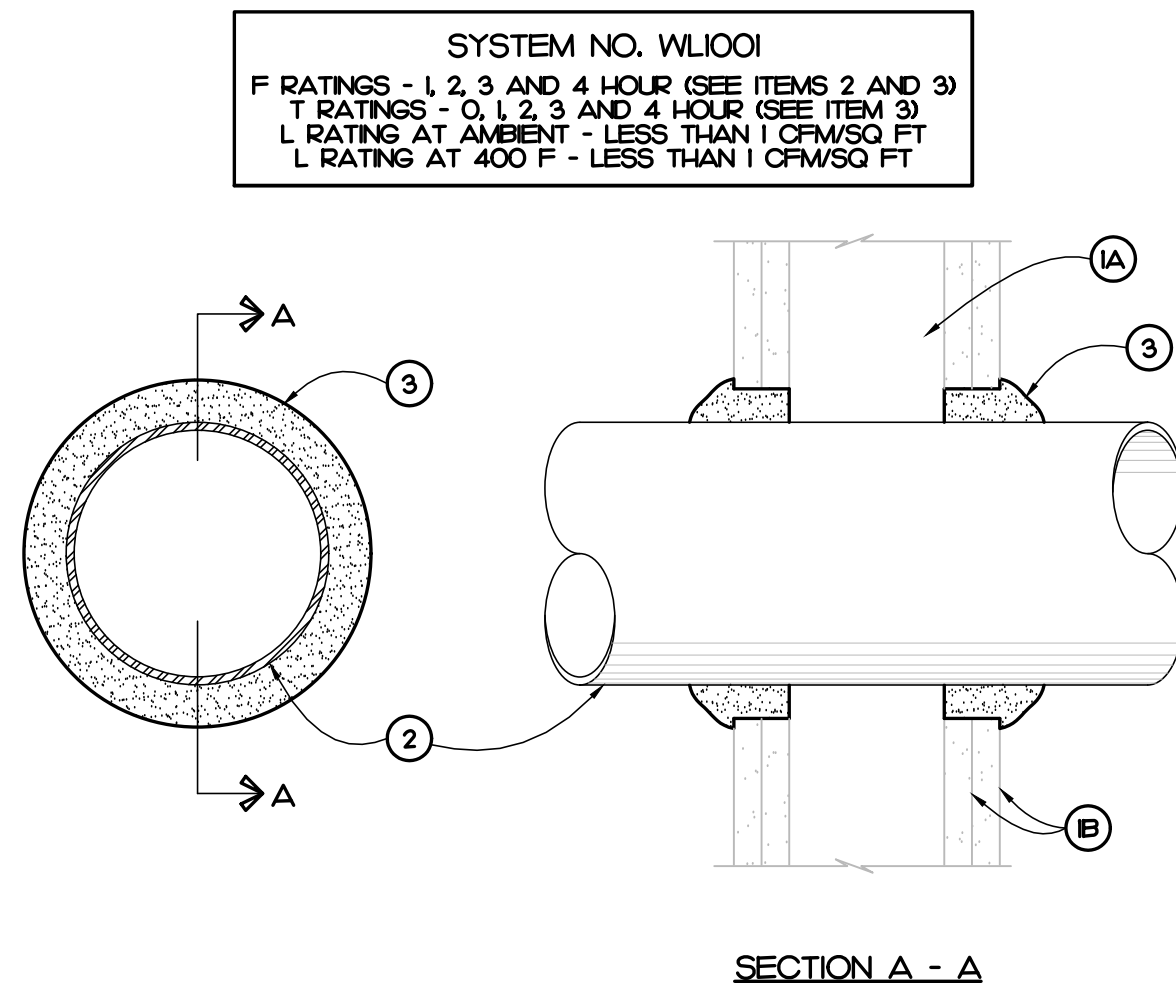
CONTENTS:
CONTROL SCHEMATICS

DATE:
AUGUST 13, 2026

DESIGNER: GWD
ENGINEER: BWF

SHEET NO.

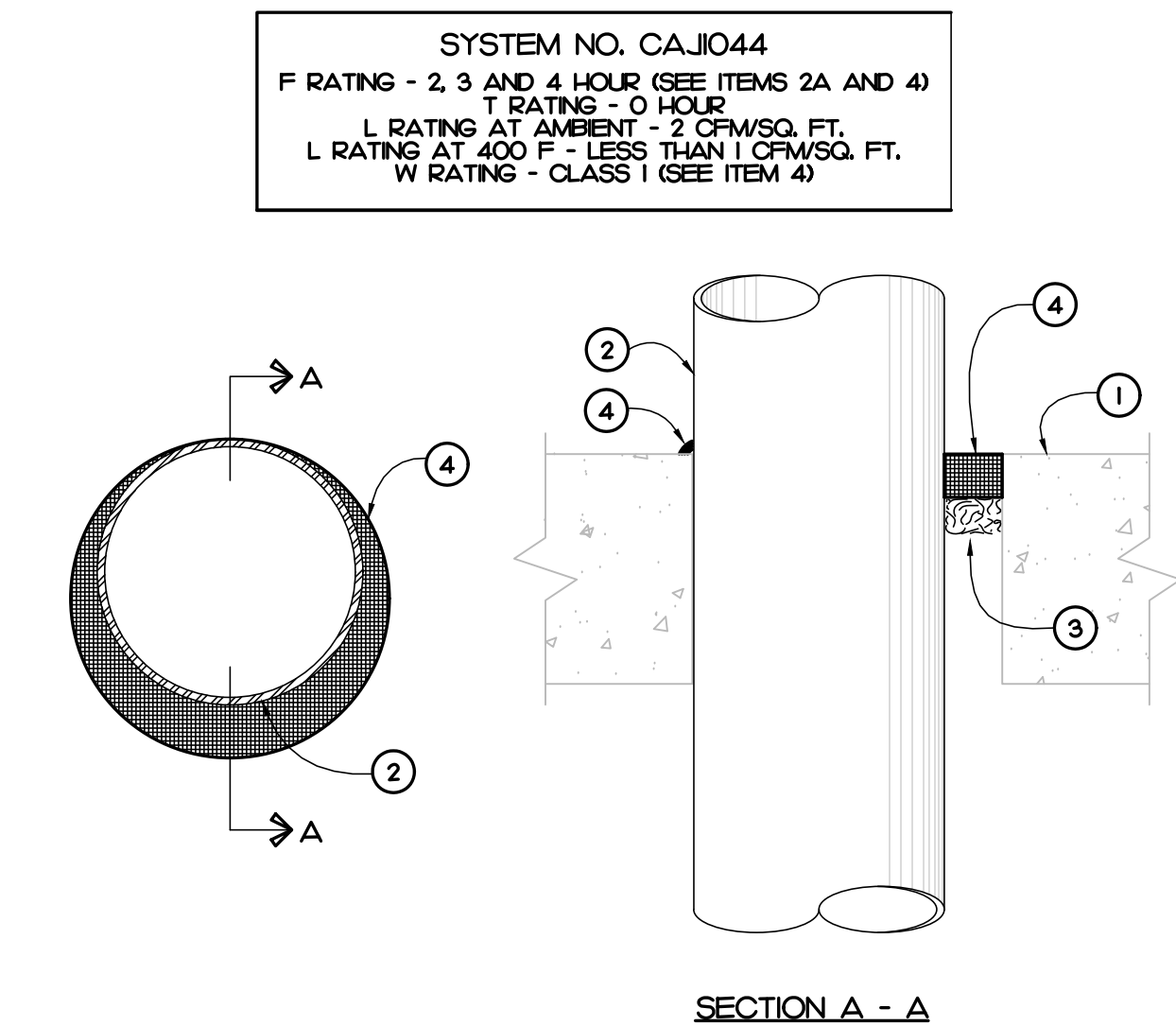
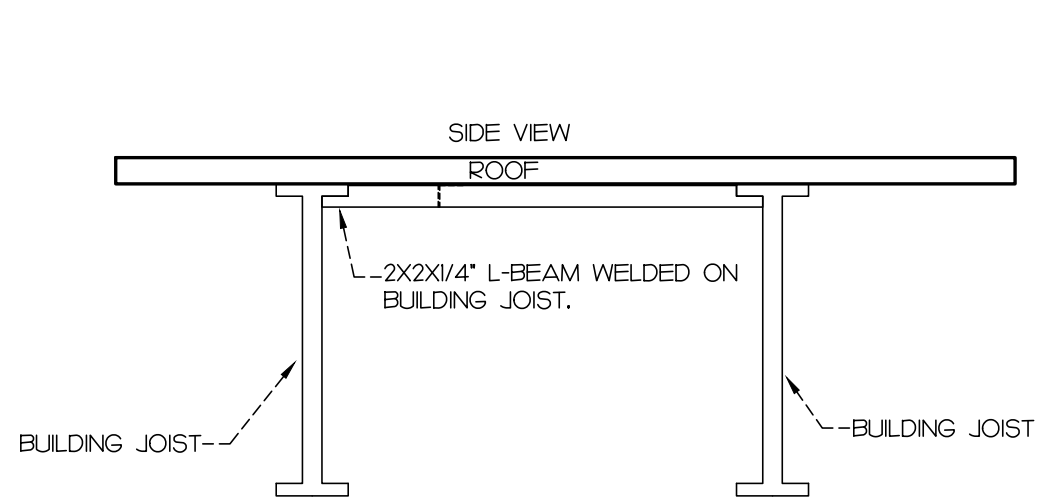
M-6



- 1 WALL ASSEMBLY - THE 1, 2, 3 OR 4 HOUR FIRE-RATED GYPSUM WALLBOARD/STUD WALL ASSEMBLY SHALL BE CONSTRUCTED OF THE MATERIALS AND IN THE MANNER DESCRIBED IN THE INDIVIDUAL U300 OR U400 SERIES WALL OR PARTITION DESIGNS IN THE UL FIRE RESISTANCE DIRECTORY AND SHALL INCLUDE THE FOLLOWING CONSTRUCTION FEATURES:
- A. STUDS - WALL FRAMING MAY CONSIST OF EITHER WOOD STUDS (MAXIMUM 2 HOUR FIRE RATED ASSEMBLY) OR STEEL CHANNEL STUDS. WOOD STUDS TO CONSIST OF NOMINAL 2" x 4" LUMBER SPACED 16" ON CENTER WITH NOMINAL 2" x 4" LUMBER END PLATES AND CROSS BRACES. STEEL STUDS TO BE MINIMUM 3 5/8" WIDE x 1 3/8" DEEP CHANNELS SPACED MAXIMUM 24" ON CENTER.
- B. GYPSUM BOARD - NOMINAL 1/2" OR 5/8" THICK, 4' WIDE WITH SQUARE OR TAPERED EDGES, THE GYPSUM WALLBOARD TYPE, THICKNESS, NUMBER OF LAYERS, FASTENER TYPE AND SHEET ORIENTATION SHALL BE AS SPECIFIED IN THE INDIVIDUAL U300 OR U400 SERIES DESIGN IN THE UL FIRE RESISTANCE DIRECTORY. MAXIMUM DIAMETER OF OPENING IS 26".
- 2 THROUGH PENETRANT - ONE METALLIC PIPE, CONDUIT OR TUBING INSTALLED EITHER CONCENTRICALLY OR ECCENTRICALLY WITHIN THE FIRESTOP SYSTEM. THE ANNULAR SPACE BETWEEN PIPE, CONDUIT OR TUBING AND PERIPHERY OF OPENING SHALL BE MINIMUM OF 0" (POINT CONTACT) TO MAXIMUM 2" PIPE, CONDUIT OR TUBING TO BE RIGIDLY SUPPORTED ON BOTH SIDES OF WALL ASSEMBLY. THE FOLLOWING TYPES AND SIZES OF METALLIC PIPES, CONDUITS OR TUBING MAY BE USED:
- A. STEEL PIPE - NOMINAL 24" DIAMETER (OR SMALLER) SCHEDULE 10 (OR HEAVIER) STEEL PIPE.
- B. IRON PIPE - NOMINAL 24" DIAMETER (OR SMALLER) SERVICE WEIGHT (OR HEAVIER) CAST IRON SOIL PIPE, NOMINAL 12" DIAMETER (OR SMALLER) OR CLASS 50 (OR HEAVIER) DUCTILE IRON PRESSURE PIPE.
- C. CONDUIT - NOMINAL 6" DIAMETER (OR SMALLER) STEEL, CONDUIT OR NOMINAL 4" DIAMETER (OR SMALLER) STEEL ELECTRICAL METALLIC TUBING.
- D. COPPER TUBING - NOMINAL 6" DIAMETER (OR SMALLER) TYPE L (OR HEAVIER) COPPER TUBING.
- E. COPPER PIPE - NOMINAL 6" DIAMETER (OR SMALLER) REGULAR (OR HEAVIER) COPPER PIPE.
- F. THROUGH PENETRATING PRODUCT* - FLEXIBLE METAL PIPING - THE FOLLOWING TYPES OF STEEL FLEXIBLE METAL GAS PIPING MAY BE USED:
1. NOMINAL 2" DIAMETER (OR SMALLER) STEEL FLEXIBLE METAL GAS PIPING, PLASTIC COVERING ON PIPING MAY OR MAY NOT BE REMOVED ON BOTH SIDES OF FLOOR OR WALL ASSEMBLY.
2. NOMINAL 1" DIAMETER (OR SMALLER) STEEL FLEXIBLE METAL GAS PIPING, PLASTIC COVERING ON PIPING MAY OR MAY NOT BE REMOVED ON BOTH SIDES OF FLOOR OR WALL ASSEMBLY.
3. NOMINAL 1" DIAMETER (OR SMALLER) STEEL FLEXIBLE METAL GAS PIPING, PLASTIC COVERING ON PIPING MAY OR MAY NOT BE REMOVED ON BOTH SIDES OF FLOOR OR WALL ASSEMBLY.
- WARD MFG INC
- 3 FILL, VOID OR CAVITY MATERIAL* - CALK OR SEALANT - MINIMUM 5/8", 1 1/4", 1 7/8" AND 2 1/2" THICKNESS OF CALK FOR 1, 2, 3 AND 4 HOUR RATED ASSEMBLIES, RESPECTIVELY, APPLIED WITHIN ANNULUS, FLUSH WITH BOTH SURFACES OF WALL. MINIMUM 1/4" DIAMETER BEAD OF CALK APPLIED TO GYPSUM BOARD/PENETRANT INTERFACE AT POINT CONTACT LOCATION ON BOTH SIDES OF WALL. THE HOURLY F RATING OF THE FIRESTOP SYSTEM IS DEPENDENT UPON THE HOURLY FIRE RATING OF THE WALL ASSEMBLY IN WHICH IT IS INSTALLED, AS SHOWN IN THE FOLLOWING TABLE. THE HOURLY T RATING OF THE FIRESTOP SYSTEM IS DEPENDENT UPON THE TYPE OR SIZE OF THE PIPE OR CONDUIT AND THE HOURLY FIRE RATING OF THE WALL ASSEMBLY IN WHICH IT IS INSTALLED, AS TABULATED BELOW:

MAXIMUM PIPE OR CONDUIT DIAMETER INCHES	F RATING HOUR	T RATING HOUR
1	1 OR 2	0, 1 OR 2
4	3 OR 4	3 OR 4
6	1 OR 2	0
12	3 OR 4	0
12	1 OR 2	0

*WHEN COPPER PIPE IS USED, T RATING IS 0 HOUR.
3M COMPANY - CP 25WB + CALK OR FB-3000 WT SEALANT.
*BEARING THE UL CLASSIFICATION MARKING

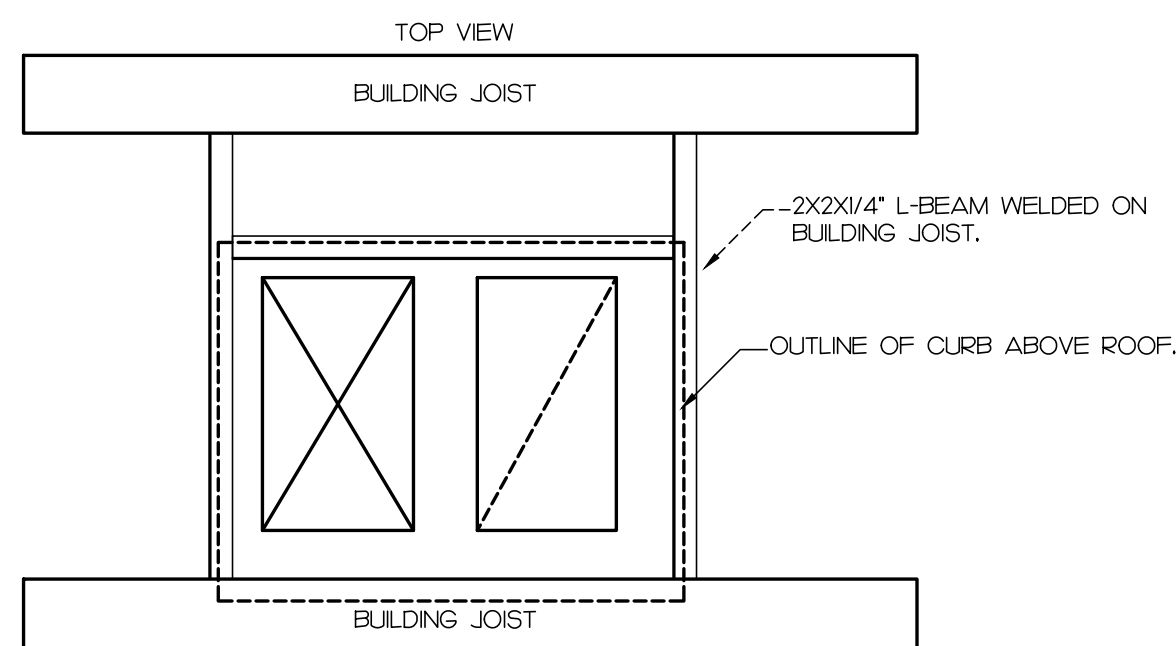


- 1 FLOOR OR WALL ASSEMBLY - LIGHTWEIGHT OR NORMAL WEIGHT (80-150 PCF) CONCRETE, EXCEPT AS NOTED IN TABLE UNDER ITEM 4, MINIMUM THICKNESS OF SOLID CONCRETE FLOOR OR WALL ASSEMBLY IS 4 1/2". FLOOR MAY ALSO BE CONSTRUCTED OF ANY MINIMUM 6" THICK UL CLASSIFIED HOLLOW CORE PRECAST CONCRETE UNITS. WHEN FLOOR IS CONSTRUCTED OF HOLLOW CORE PRECAST CONCRETE UNITS, PACKING MATERIAL (ITEM 3) AND CALK/FILL MATERIAL (ITEM 4) TO BE INSTALLED SYMMETRICALLY ON BOTH SIDES OF FLOOR, FLUSH WITH FLOOR SURFACE. WALL ASSEMBLY MAY ALSO BE CONSTRUCTED OF ANY UL CLASSIFIED CONCRETE BLOCKS. MAXIMUM DIAMETER OF OPENING IS IN SOLID LIGHTWEIGHT OR NORMAL WEIGHT CONCRETE, FLOOR IS 32" MAXIMUM DIAMETER OF OPENING IN FLOOR CONSTRUCTED OF HOLLOW-CORE PRECAST CONCRETE UNITS IS 7". SEE CONCRETE BLOCKS (CAZT) AND PRECAST CONCRETE UNITS (CFTV) CATEGORIES IN THE FIRE RESISTANCE DIRECTORY FOR NAMES OF MANUFACTURERS.
- 2A STEEL SLEEVE (OPTIONAL, NOT SHOWN) - MAXIMUM 5" ID (OR SMALLER) SCHEDULE 10 (OR HEAVIER) STEEL SLEEVE CAST OR GROUTED INTO FLOOR OR WALL ASSEMBLY. SLEEVE MAY EXTEND A MAXIMUM OF 2" ABOVE TOP OF FLOOR OR BEYOND EITHER SURFACE OF WALL. MAXIMUM 16" ID (OR SMALLER) MINIMUM 0.025" WALL THICKNESS (OR HEAVIER) GALVANIZED STEEL SLEEVE CAST OR GROUTED INTO FLOOR OR WALL ASSEMBLY. SLEEVE MAY EXTEND A MAXIMUM OF 1/2" BEYOND EITHER SURFACE OF FLOOR OR WALL.
- 2 THROUGH PENETRANTS - ONE METALLIC PIPE, CONDUIT OR TUBING TO BE INSTALLED EITHER CONCENTRICALLY OR ECCENTRICALLY WITHIN THE FIRESTOP SYSTEM. MAXIMUM ANNULAR SPACE BETWEEN PIPE, CONDUIT OR TUBING AND EDGE OF THROUGH OPENING OR SLEEVE IS DEPENDENT ON THE PARAMETERS SHOWN IN ITEM 4. MINIMUM ANNULAR SPACE BETWEEN PIPE OR CONDUIT AND EDGE OF THROUGH OPENING IS 0" (POINT CONTACT). PIPE, CONDUIT OR TUBING TO BE RIGIDLY SUPPORTED ON BOTH SIDES OF FLOOR OR WALL ASSEMBLY. THE FOLLOWING TYPES AND SIZES OF METALLIC PIPES, CONDUITS OR TUBING MAY BE USED:
- A. STEEL PIPE - NOMINAL 30" DIAMETER (OR SMALLER) SCHEDULE 10 (OR HEAVIER) STEEL PIPE.
- B. IRON PIPE - NOMINAL 30" DIAMETER (OR SMALLER) CAST OR DUCTILE IRON PIPE.
- C. CONDUIT - NOMINAL 6" DIAMETER (OR SMALLER) RIGID STEEL CONDUIT.
- D. CONDUIT - NOMINAL 4" DIAMETER (OR SMALLER) STEEL ELECTRICAL METALLIC TUBING.
- E. COPPER - TUBING NOMINAL 6" DIAMETER (OR SMALLER) TYPE L (OR HEAVIER) COPPER TUBE.
- F. COPPER PIPE - NOMINAL 6" DIAMETER (OR SMALLER) REGULAR (OR HEAVIER) COPPER PIPE.
- 3 PACKING MATERIAL - POLYETHYLENE BACKER ROD OR NOMINAL 1" THICKNESS OF TIGHTLY-PACKED MINERAL WOOL BATT OR GLASS FIBER INSULATION FIRMLY PACKED INTO OPENING AS A PERMANENT FORM. PACKING MATERIAL TO BE RECESSED FROM TOP SURFACE OF FLOOR OR FROM BOTH SURFACES OF WALL AS REQUIRED TO ACCOMMODATE THE REQUIRED THICKNESS OF CALK/FILL MATERIAL (ITEM 4).
- 4 FILL, VOID OR CAVITY MATERIAL* - CALK OR SEALANT - APPLIED TO FILL THE ANNULAR SPACE FLUSH WITH TOP SURFACE OF FLOOR. IN WALL ASSEMBLIES, REQUIRED CALK THICKNESS TO BE INSTALLED SYMMETRICALLY ON BOTH SIDES OF WALL, FLUSH WITH WALL SURFACE, AT POINT CONTACT LOCATION BETWEEN PENETRANT AND SLEEVE OR BETWEEN PENETRANT AND CONCRETE. A MINIMUM 1/4" DIAMETER BEAD OF CALK SHALL BE APPLIED AT TOP SURFACE OF FLOOR AND AT BOTH SURFACES OF WALL. THE HOURLY F RATING AND THE MINIMUM REQUIRED CALK THICKNESSES ARE DEPENDENT UPON A NUMBER OF PARAMETERS, AS SHOWN IN THE FOLLOWING TABLE:

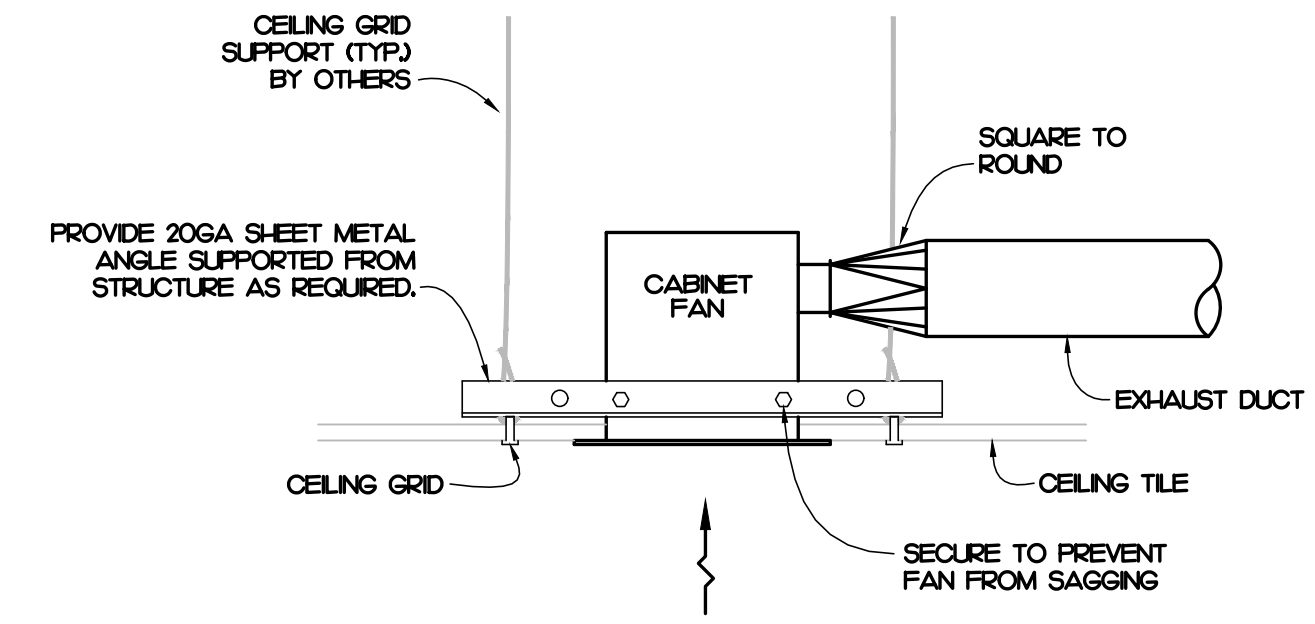
MINIMUM FLOOR OR WALL THICKNESS INCHES	NOMINAL PIPE TUBE OR CONDUIT DIAMETER INCHES	MAXIMUM ANNULAR SPACE INCHES	MINIMUM CALK THICKNESS INCHES	F RATING
2 1/2	1/2 - 12	1 3/8	1/2	2
2 1/2	1/2 - 12	1 3/8	1/2	2
4 1/2	1/2 - 6	1 3/8	1/4(0)	2
4 1/2	1/2 - 12	1 1/4	1/2	3
4 1/2	1/2 - 20	2	1	3
4 1/2	1/2 - 20	2	1	3
4 1/2	1/2 - 12	3 1/4	1	3
4 1/2	22 - 30	2	2	3
5 1/2	1/2 - 6	1 3/8	1 (0)	4

(a) MINIMUM 2" THICKNESS OF MINERAL WOOL BATT INSULATION REQUIRED IN ANNULAR SPACE.
(b) MINIMUM 1" THICKNESS OF MINERAL WOOL BATT INSULATION REQUIRED IN ANNULAR SPACE ON BOTH SIDES OF FLOOR OR WALL ASSEMBLY. MINIMUM 1" THICKNESS OF CALK TO BE INSTALLED FLUSH WITH EACH SURFACE OF FLOOR OR WALL ASSEMBLY.

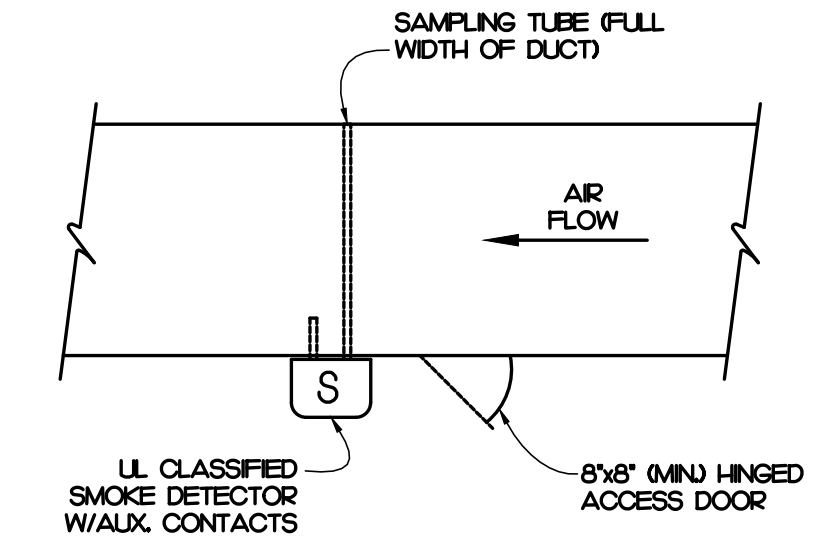
3M COMPANY - CP 25WB + CALK OR FB-3000 WT SEALANT. (THE W RATING APPLIES ONLY WHEN FB-3000 WT SEALANT IS USED).
*BEARING THE UL CLASSIFICATION MARKING



RTU-2 AND RTU-3
07 - CURB FRAMING DETAIL
SCALE: N.T.S.

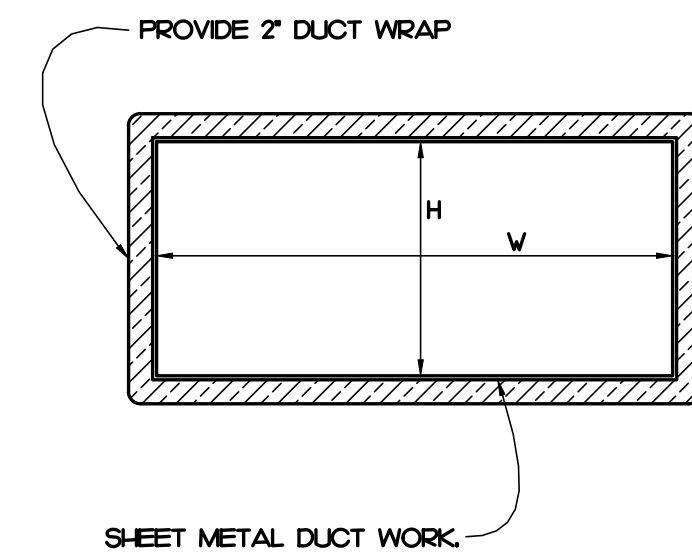


06 - CABINET FAN DETAIL
SCALE: N.T.S.



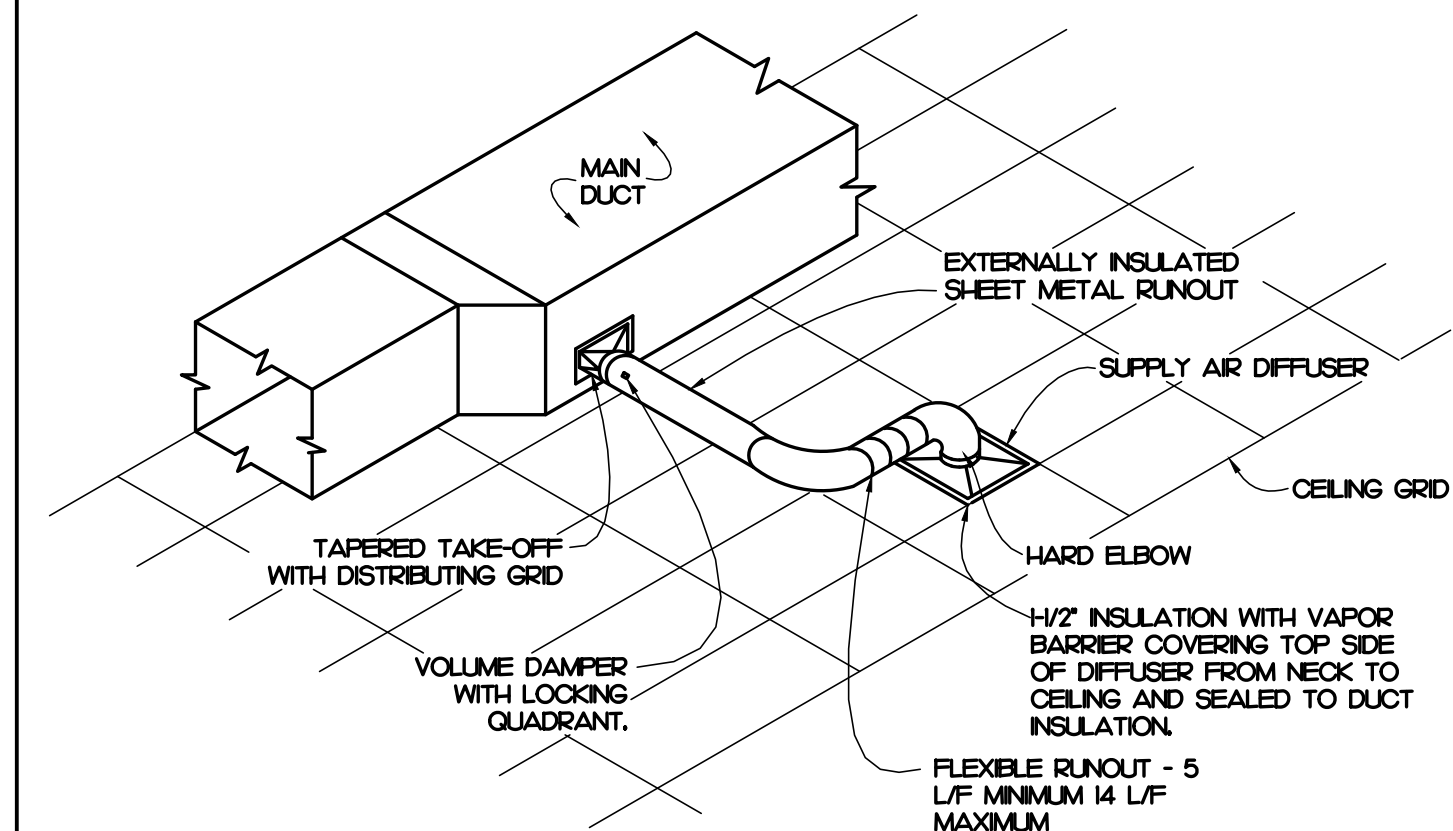
- NOTES:
- 1 IF DUCT WIDTH EXCEEDS 36", SAMPLING TUBE SHALL PENETRATE FAR SIDE OF DUCT & HAVE INTERMEDIATE SUPPORT. PENETRATION SHALL BE SEALED AIRTIGHT.
- 2 LOCATE SMOKE DETECTOR IN RETURN DUCT UPSTREAM OF OUTSIDE AIR INTAKE IN NON-TURBULENT AIRSTREAM. INSTALL PER MFG'S DETAILED INSTALLATION INSTRUCTIONS.

05 - DUCT DETECTOR DETAIL
SCALE: N.T.S.

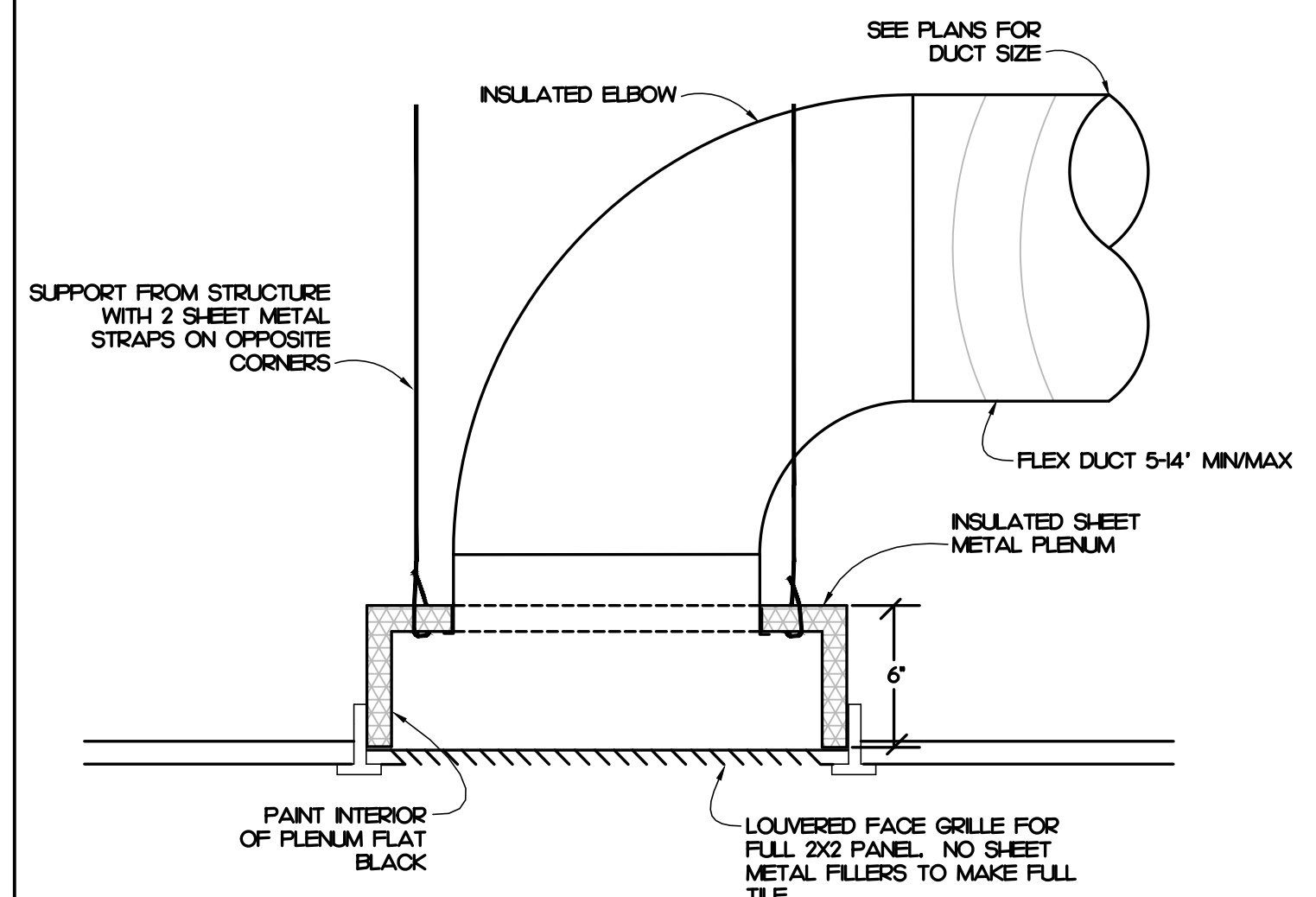


- NOTES:
- DIMENSIONS ARE INSIDE CLEAR WIDTH X HEIGHT
- APPLY MASTIC TO ALL JOINTS

04 - DUCT DETAIL
SCALE: N.T.S.

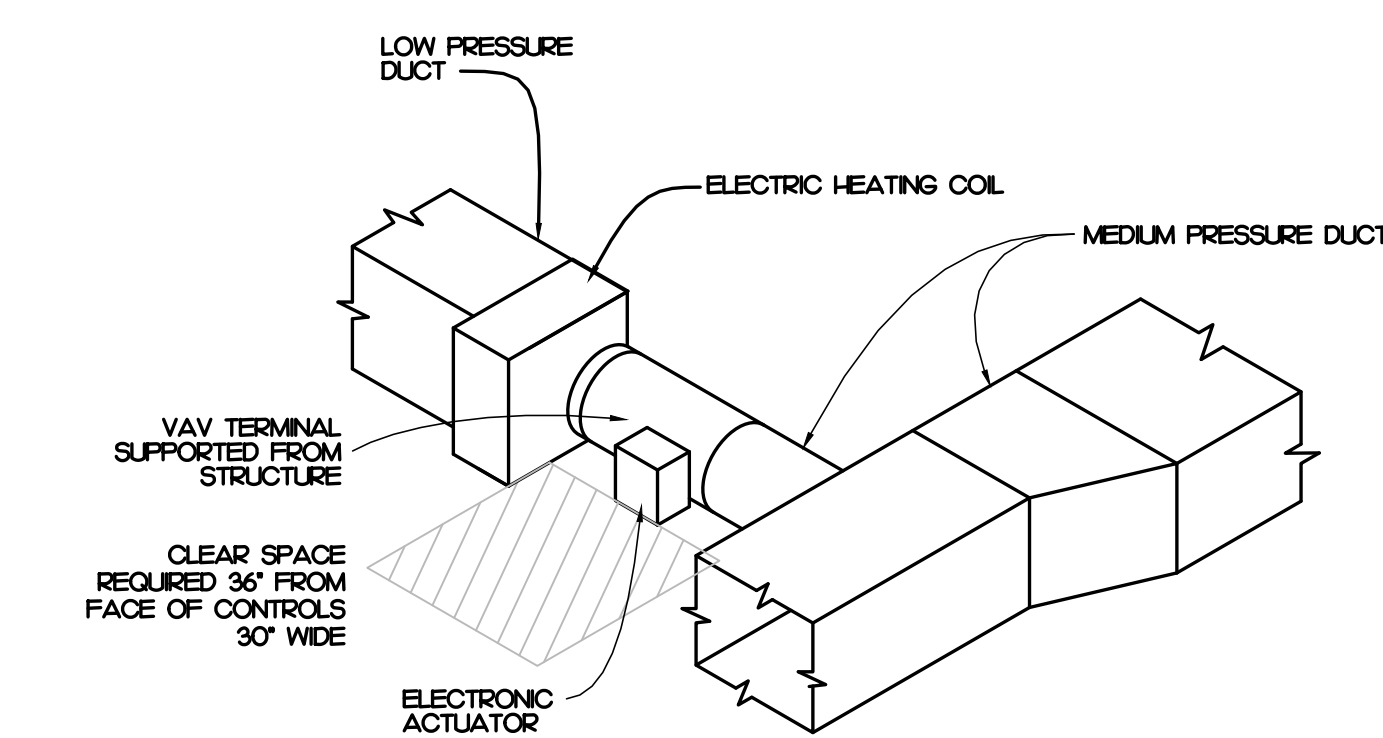


03 - SUPPLY GRILLE DETAIL
SCALE: N.T.S.



02 - RETURN GRILLE DETAIL
SCALE: N.T.S.

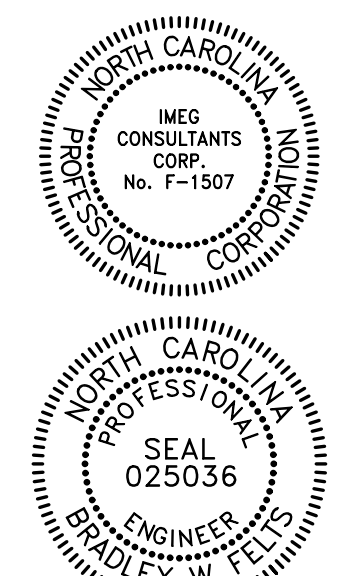
- NOTE:
- PROVIDE 36" CLEAR IN FRONT OF CONTROL BOX.
- PROVIDE 16 DUCT DIAMETERS OF STRAIGHT DUCT INTO INLET OF TERMINAL.



01 - VAV DETAIL
SCALE: N.T.S.

REVISIONS					

PROJECT NO.:
25007326.01



8/13/2026
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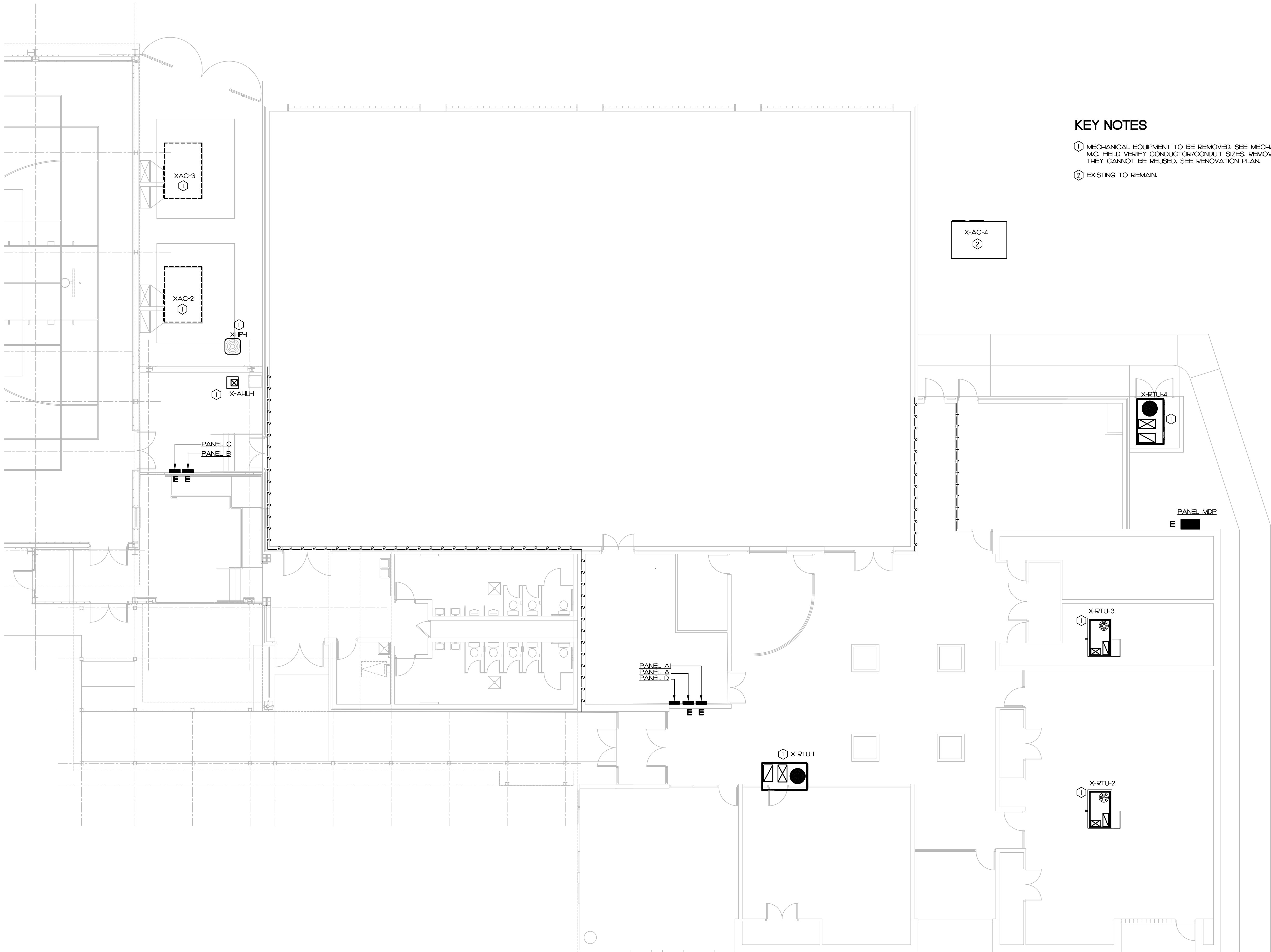
HVAC RENOVATION
GREEN ROAD COMMUNITY CENTER GYM
4201 GREEN ROAD
RALEIGH
NORTH CAROLINA

CONTENTS:
MECHANICAL DETAIL

DATE:
AUGUST 13, 2026

DESIGNER: GWD
ENGINEER: BWF

SHEET NO.
M-7



KEY NOTES

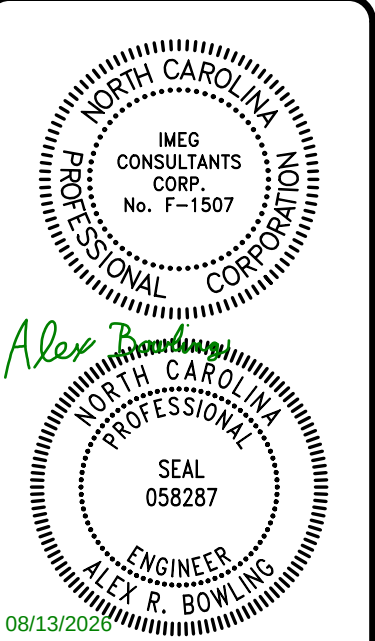
- ① MECHANICAL EQUIPMENT TO BE REMOVED, SEE MECHANICAL PLANS. FIELD COORDINATE WORK WITH M.C. FIELD VERIFY CONDUCTOR/CONDUIT SIZES. REMOVE CONDUCTOR/CONDUITS BACK TO PANEL IF THEY CANNOT BE REUSED. SEE RENOVATION PLAN.
- ② EXISTING TO REMAIN.

01 - **DEMOLITION PLAN - ELECTRICAL**

SCALE: 1/8" = 1'-0"

REVISIONS	

PROJECT NO.:
25007326.01



3221 BLUE RIDGE ROAD, SUITE 113
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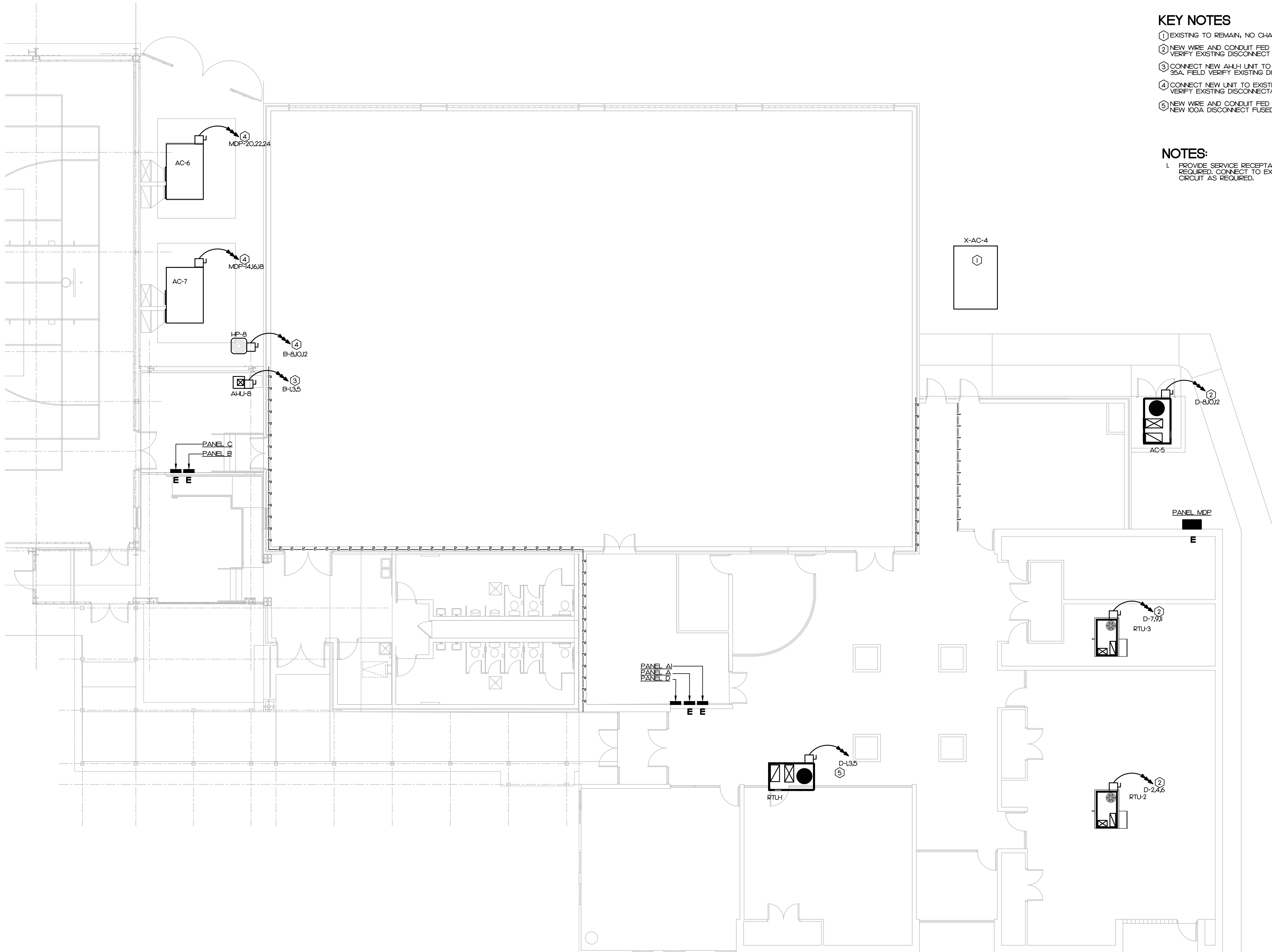
HVAC RENOVATION
GREEN ROAD COMMUNITY CENTER GYM
4201 GREEN ROAD
RALEIGH **NORTH CAROLINA**

CONTENTS:
ELECTRICAL
DEMOLITION PLAN

DATE:
AUGUST 13, 2026

DESIGNER: **PEQ**
ENGINEER: **ARB**

SHEET NO.
E-2



KEY NOTES

- ① EXISTING TO REMAIN, NO CHANGE, SHOWN FOR REFERENCE.
- ② NEW WIRE AND CONDUIT FED FROM NEW PANEL D. FIELD VERIFY EXISTING DISCONNECT SIZE IS ADEQUATE.
- ③ CONNECT NEW AHU-1 UNIT TO EXISTING DISCONNECT, FUSE AT 35A. FIELD VERIFY EXISTING DISCONNECT SIZE IS ADEQUATE.
- ④ CONNECT NEW UNIT TO EXISTING DISCONNECT CIRCUIT, FIELD VERIFY EXISTING DISCONNECT/FUSE SIZE IS ADEQUATE.
- ⑤ NEW WIRE AND CONDUIT FED FROM NEW PANEL D. NEW 100A DISCONNECT FUSED AT 90 AMPS.

NOTES:

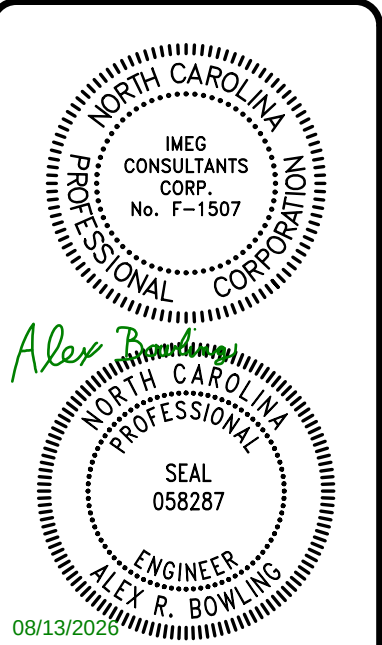
1. PROVIDE SERVICE RECEPTACLE PER NEC 210.63 AS REQUIRED, CONNECT TO EXISTING ROOFTOP RECEPTACLE CIRCUIT AS REQUIRED.

01 - RENOVATION PLAN - ELECTRICAL

SCALE: 1/8" = 1'-0"

REVISIONS	

PROJECT NO.:
25007326.01



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PH: (919) 571-1111

HVAC RENOVATION
GREEN ROAD COMMUNITY CENTER GYM
4201 GREEN ROAD
RALEIGH NORTH CAROLINA

CONTENTS:
ELECTRICAL
RENOVATION PLAN

DATE:
AUGUST 13, 2026

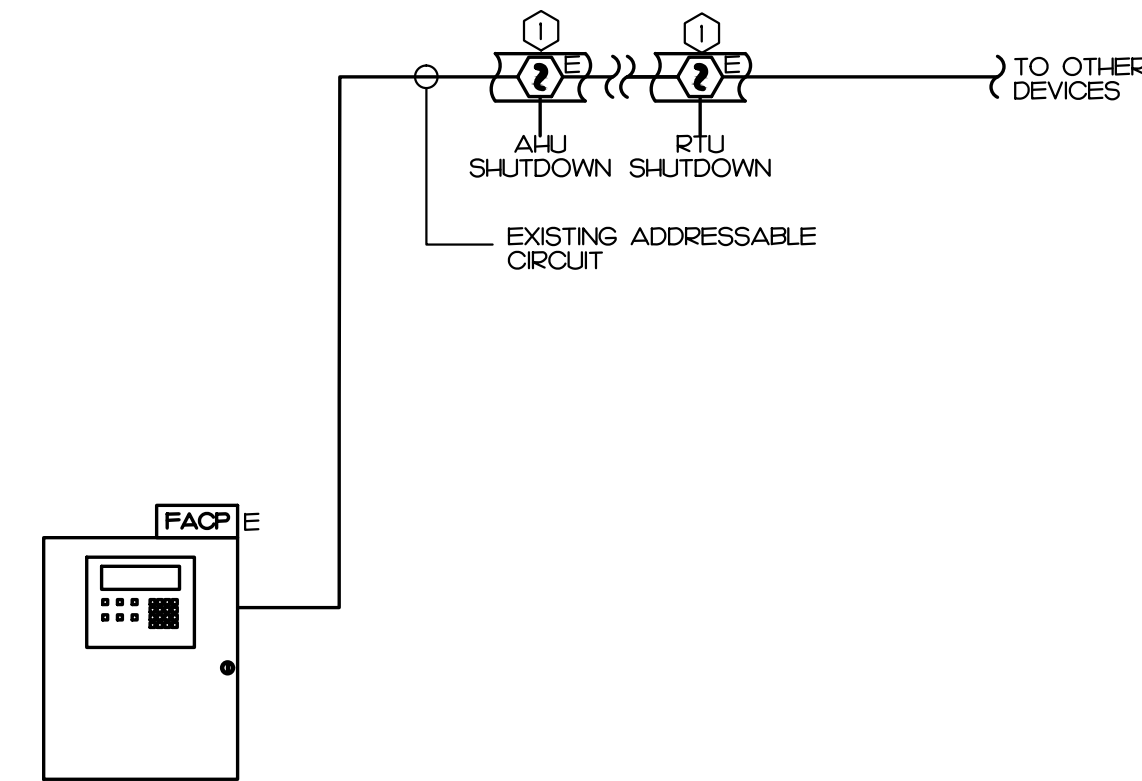
DESIGNER: PEG
ENGINEER: ARB

SHEET NO.

E-3



SYMBOL	DESCRIPTION	REMARKS
	EXISTING: DUOT SMOKE DETECTOR, PHOTOELECTRIC WITH SAMPLING TUBE AND REMOTE TEST SWITCH. SEE NOTE ON PLAN.	EXISTING
E	EXISTING: FIRE ALARM CONTROL PANEL, SURFACE MOUNTED, ADDRESSABLE.	HONEYWELL VISTA 128 FFP
E	EXISTING: FIRE ALARM REMOTE ANNUNCIATOR, SURFACE MOUNTED, SHOWN FOR REFERENCE, NO WORK.	EXISTING



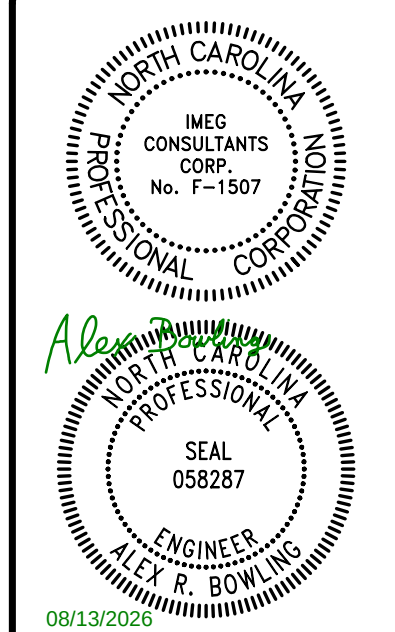
① TYPICAL AT EXISTING AHU/RTU TO BE REPLACED.
DISCONNECT SHUTDOWN CIRCUIT FROM EXISTING
AHU/RTU AND RECONNECT TO THE REPLACED AHU/RTU
AS REQUIRED.

1. SEE PLANS FOR QUANTITY AND LOCATION OF ALL EQUIPMENT.
 - THIS PLAN SHOWS ONLY ADDITIONAL EQUIPMENT TO EXISTING FIRE ALARM SYSTEM.
 - ELECTRICAL SHALL BE RESPONSIBLE FOR EXISTING SYSTEM PRIOR TO ANY MODIFICATION.
 - ALL MODIFICATION SHALL NOT VOID EXISTING SYSTEM CURRENT WARRANTY OR SERVICE CONTRACT.
 - REPROGRAM EXISTING PANEL AS REQUIRED BY E.C.
 - SCOPE OF WORK IS LIMITED ONLY TO THE FIRE ALARM DEVICES THAT ARE ASSOCIATED WITH THE HVAC EQUIPMENT.
2. CONTRACTOR SHALL PROVIDE COMPLETE DOCUMENT PER 2018 FIRE CODE SECTION 907J AND 90712 TO ENGINEER FOR APPROVAL PRIOR TO SUBMIT TO AND TESTING BY THE CITY OF RALEIGH FIRE MARSHAL'S OFFICE.
3. PLACEARD THE ENTIRE FIRE ALARM SYSTEM. PROVIDE PANEL AND CIRCUIT NUMBERS ON A NAME PLATE AFFIXED TO THE FACE OF THE FIRE ALARM CONTROL PANEL.
4. CONTRACTOR SHALL PROVIDE ZONE MAPS COMPLETE WITH ADDRESSES FOR EACH FIRE ALARM DEVICE IN WOODEN FRAME ADJACENT TO THE EXISTING FIRE ALARM CONTROL PANEL.
5. ELECTRICAL CONTRACTOR SHALL PROVIDE BATTERY CALCULATIONS AND CUT SHEETS FOR FIRE ALARM SYSTEM TO ENGINEER FOR APPROVAL.
6. ALL WIRING SHALL BE SUPERVISED.
7. ALL WIRING SHALL BE PER MANUFACTURER'S SPECIFICATIONS.
8. ALL WIRING IN WALLS OR FURRED SPACES SHALL BE IN CONDUIT.
9. WHERE PERMITTED BY CODE, WIRING ABOVE ACCESSIBLE CEILINGS MAY BE RUN EXPOSED AND THE FOLLOWING REQUIREMENTS SHALL BE MET:
 - WIRING SHALL BE PLENUM RATED WHERE APPLICABLE.
 - PROVIDE BRIDLE RINGS FOR INDEPENDENT FIRE ALARM CABLE SUPPORT UNLESS SPECIFICALLY NOTED OTHERWISE. ANALOG LOOP WIRING INCOMING AND OUTGOING SHALL NOT BE SUPPORTED IN THE SAME BRIDLE RING.
10. ADDRESSABLE SLC CIRCUIT REQUIREMENTS:
 - WIRING CLASS SHALL BE PER EXISTING. FIELD VERIFY.
11. FOR ALL AHU UNITS WITH REQUIRED DUCT DETECTOR PER MECHANICAL PLAN.
 - DUCT DETECTOR ARE EXISTING
 - FIRE ALARM CONTRACTOR SHALL PROVIDE WIRING AND RELAYS AS REQUIRED FOR AIR HANDLING SHUTDOWN. FIELD COORDINATE AIR HANDLING UNIT SHUTDOWN WITH MECHANICAL. THE FIRE ALARM FINAL CONNECTIONS OF WIRING FOR HVAC SYSTEM SHALL BE BY THE ELECTRICAL CONTRACTOR.
 - CONTRACTOR SHALL PROVIDE A SMOKE MACHINE TO TEST THE DUCT DETECTION SECTION OF THE FIRE ALARM SYSTEM. BLOWERS AND/OR MAGNETS FOR TESTING OF THE DUCT DETECTION SYSTEM IS PROHIBITED.

[illegible]

REVISIONS					

PROJECT NO.:
25007326.01



3221 BLUE RIDGE ROAD, SUITE 113
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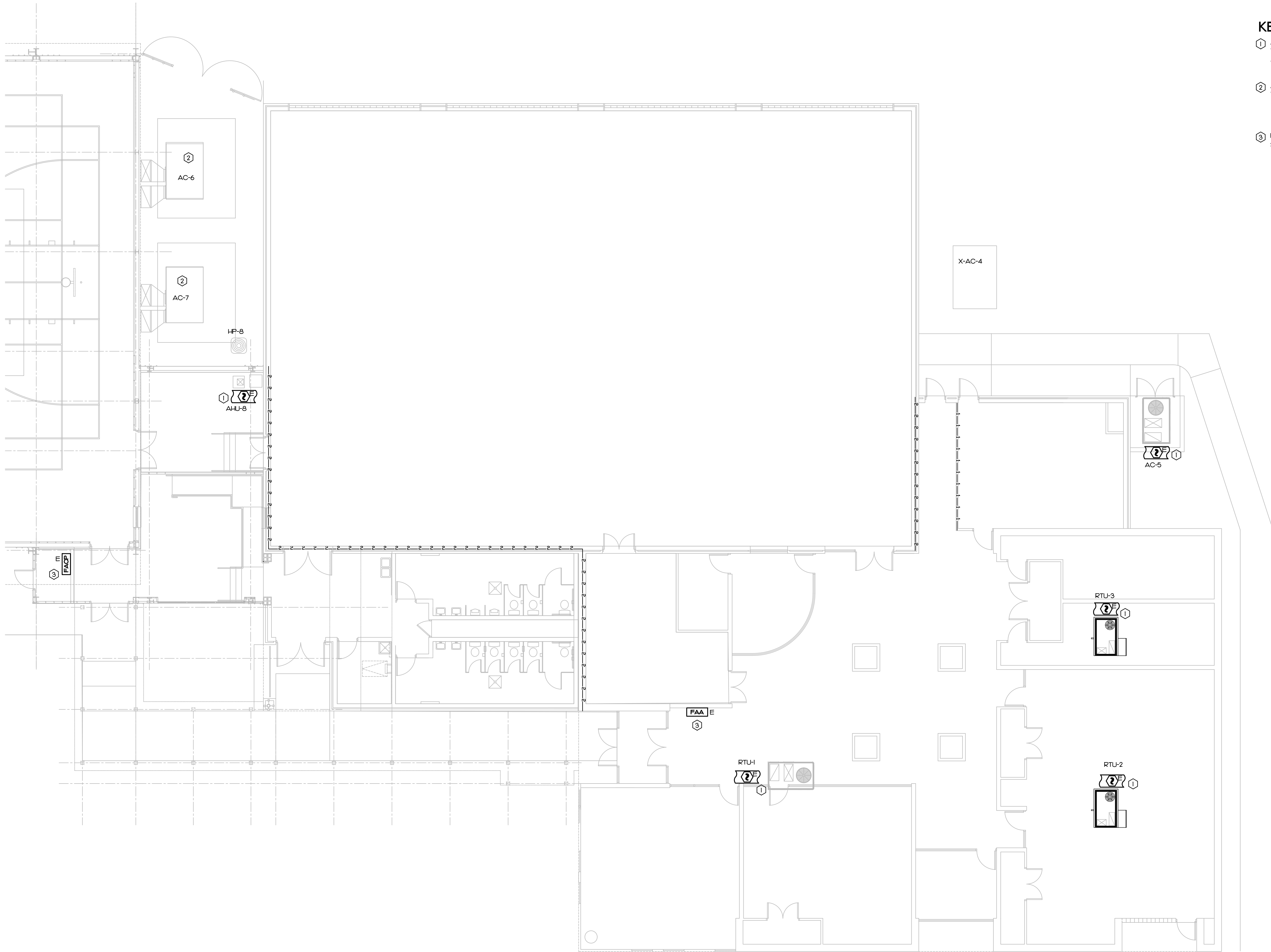
CONTENTS:
FIRE ALARM LEGEND,
NOTES, DETAILS

DATE:
AUGUST 13, 2026

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ENGINEER: ARB

SHEET NO.

FA-1



KEY NOTES

- ① AT EXISTING UNIT TO BE REPLACED:
- REMOVE AND REINSTALL DUCT DETECTOR PER M.C. INSTRUCTION.
 - REMOVE EXISTING SHUTDOWN CIRCUITRY FROM EXISTING UNIT AND RECONNECT TO NEW REPLACED UNIT AS REQUIRED.
- ② AT EXISTING UNIT TO BE REPLACED:
- E.C. SHALL FIELD VERIFY EXISTING FIRE ALARM SHUTDOWN CONNECTION. IF THERE IS AN EXISTING FIRE ALARM SHUTDOWN CONNECTION, DISCONNECT FROM EXISTING UNIT AND CONNECT TO NEW REPLACED UNIT AS REQUIRED.
- ③ E.C. SHALL PROVIDE NEW ZONE MAP AT EXISTING FA-2 AND FIRE ALARM REMOTE ANNUNCIATOR. PER SURVEY, THERE IS NO FIRE ALARM ZONE MAP IN THE BUILDING.

REVISIONS	

PROJECT NO.:
25007326.01

IMEG CONSULTANTS CORP.
No. F-1507
PROFESSIONAL CORPORATION

Alexander
ALEXANDER R. BOWLING
SEAL 058287
ENGINEER
08/13/2026

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HVAC RENOVATION
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CONTENTS:
FIRE ALARM
RENOVATION PLAN

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FA-2

01 - RENOVATION PLAN - FIRE ALARM

SCALE: 1/8" = 1'-0"